



PEDIATRIC MOONSHOT

Pediatric Cardiac Atlas **The Complete Pediatric Cardiac Defect and Disease Challenge**

104 Challenges

AI can accelerate diagnosis and treatment for children rurally, locally and globally

“If you know the enemy and know yourself, you need not fear the result of a hundred battles.”
— Sun Tzu, The Art of War

To defeat pediatric heart disease, we must first define it. The Pediatric Cardiology Atlas catalogs 104 distinct cardiac conditions affecting children worldwide, spanning congenital structural defects, cardiomyopathies, arrhythmias, inflammatory diseases, vascular disorders, and rare cardiac tumors. For each condition, the atlas details its biology, U.S. and global incidence, diagnostic workup, and treatment — distinguishing established standard-of-care protocols from emerging investigational therapies. A significant proportion carry hereditary predisposition, classified as Defining, Dominant, or Associated based on the strength of the germline contribution.

Diagnosis is rarely straightforward. Congenital defects may be detected on fetal ultrasound, discovered on a newborn exam, or remain hidden until a child collapses during exercise. Cardiomyopathies present with nonspecific symptoms — fatigue, poor feeding, shortness of breath — easily attributed to more common conditions. Inherited arrhythmias are particularly treacherous: the heart appears structurally normal until sudden cardiac arrest becomes the first sign of disease. Rare conditions carry the additional burden of diagnostic odyssey, families cycling through years of inconclusive workups before the correct diagnosis is reached.

Standard-of-care spans surveillance, medical management, catheter-based intervention, and surgical repair. Conditions that were uniformly fatal a generation ago now carry survival rates into adulthood. Emerging therapies are advancing rapidly: gene therapy trials for inherited cardiomyopathies, myosin modulators and SGLT2 inhibitors for pediatric heart failure, transcatheter valve technologies, and precision agents targeting RASopathy and channelopathy pathways.

Yet structural challenges persist. 86% of the rural counties in the United States have pediatric cardiologist. Patient populations are small and scattered, making evidence thin and trials slow to enroll. Genetic risk cascades through families but screening remains inconsistent. Children who survive complex repairs face lifelong surveillance needs that current systems track poorly.

AI can accelerate diagnosis and treatment of cardiac diseases. Pediatric cardiac disease can end with AI.

Pediatric Cardiac Conditions: Complete Clinical Reference

104 Conditions | 11 Categories | Version 9 —

Hereditary Predisposition:  Defining (condition IS a germline syndrome)  Dominant (germline mutation in >25% of cases)  Associated (strong syndromic link; not universal)

STRUCTURAL CONGENITAL HEART DISEASE					
#	CONDITION	DESCRIPTION	FREQUENCY	DIAGNOSIS	TREATMENT
1	Bicuspid Aortic Valve 	Two leaflets instead of three on the aortic valve. Leads to stenosis, regurgitation, and aortic root dilation over decades; often asymptomatic until adulthood.	1-2% of population (~3-6M Americans); most common CHD overall. Males 2-3x more affected. US: ~3.3M–4.4M total; ~1.4M–2.2M pediatric/adolescent Global: ~100M–150M (all ages) Source: Siu & Silversides, JACC 2010; BAV prevalence 1–2% all ages	Echocardiography (TTE/TEE); CT/MRI for aortic sizing. Often incidentally found.	SOC: Surveillance echocardiography until intervention thresholds met. Surgical valve repair or replacement; Ross procedure preferred in children and young adults. TAVR for adult patients with suitable anatomy and prior surgical/biological failure. Emerging: Transcatheter valve-in-valve TAVR for failed bioprostheses; NOTCH1/ROBO4 genotype-guided aortic surveillance protocols; aortic root remodeling techniques to delay replacement.
2	Ventricular Septal Defect (VSD)	Hole in the wall separating the two lower chambers. Blood shunts left to right, overloading the lungs. Small VSDs often close spontaneously.	3-4 per 1,000 live births; 20-30% of all CHD. Most common CHD diagnosed at birth. US: ~10,800–14,400 new/yr; ~250,000–350,000 pediatric prevalent Global: ~420,000–560,000 new/yr Source: van der Linde et al., JACC 2011; Liu et al., IJE 2019; 3–4/1,000 live births	Echocardiography. Prenatal fetal echo. Cardiac catheterization for complex cases.	SOC: Observation for small VSDs (70–80% close spontaneously). Surgical patch closure for large or symptomatic VSDs. Transcatheter device closure for muscular VSDs (Amplatzer Muscular VSD Occluder). Emerging: Transcatheter perimembranous VSD closure: Lifetech KONAR-MF device (CE-marked; US trials ongoing). Perventricular hybrid muscular VSD closure via mini-sternotomy.
3	Atrial Septal Defect (ASD)	Hole in the wall separating the upper chambers. Low-pressure shunt causes right heart enlargement	1-2 per 1,000 live births; ~13% of all CHD. Up to 30% undiagnosed until	Echocardiography (TTE/TEE). Bubble study for PFO differentiation. Cardiac MRI for complex	SOC: Transcatheter device closure (Amplatzer Septal Occluder, Gore Cardioform) for secundum ASDs — >80% of cases. Surgical closure for primum, sinus

		and eventual pulmonary hypertension if untreated.	after age 30. US: ~3,600–7,200 new/yr; ~150,000–300,000 pediatric prevalent Global: ~140,000–280,000 new/yr Source: Liu et al., IJE 2019; ~1–2/1,000 live births; prevalence increased 6-fold with echo era	anatomy.	venosus, or complex defects. Emerging: Transcatheter closure of sinus venosus ASDs with partial anomalous pulmonary venous return (Gore ASSURED device — US pivotal trial completed). Fenestrated closure devices for borderline pulmonary hypertension.
4	Patent Ductus Arteriosus (PDA)	Fetal vessel connecting pulmonary artery to aorta fails to close after birth. Blood shunts aorta to pulmonary, overloading lungs.	1 in 2,000 full-term births; ~70% of infants born <28 wks. 2-3x mortality risk in VLBW infants. US: ~1,800–3,600 new full-term cases/yr; ~100,000–200,000 premature-related/yr Global: ~70,000–140,000 new full-term cases/yr Source: van der Linde et al., JACC 2011; 1/2,000 full-term births; ~70% of infants <28 wks	Echocardiography; continuous murmur, bounding pulses, widened pulse pressure.	SOC: Indomethacin or ibuprofen (medical closure, 70–80% in premature infants). Transcatheter closure: Amplatzer Piccolo Occluder (FDA 2019, down to 700g) for hemodynamically significant PDAs. Surgical ligation for refractory cases. Emerging: Acetaminophen as alternative medical closure agent. Catheter-based closure in extremely premature infants <1 kg at specialized centers. Conservative/expectant management strategies for very premature infants gaining evidence.
5	Pulmonary Stenosis 🧬	Narrowing at the pulmonary valve or surrounding structures. Three subtypes: (1) Valvar PS (90%); (2) Subvalvar/Infundibular PS -- muscular/fibrous obstruction, requires surgical resection; (3) Supravalvar/Branch PA stenosis -- associated with Williams, Noonan, Alagille	~8-10% of all CHD; ~8 per 10,000 live births. Valvar PS: ~90% of cases. Mild valvar PS (gradient <40 mmHg): ~60% of valvar cases. US: ~2,900–3,600 new/yr; ~120,000–150,000 pediatric prevalent Global: ~112,000–	Echo with Doppler gradient (distinguish valvar, subvalvar, supravalvar). Cardiac catheterization for intervention planning. CT/MRI for complex anatomy or supravalvar disease.	SOC: Valvar PS: balloon valvuloplasty for gradient >40 mmHg (>90% success). Surgical repair for dysplastic valves (Noonan syndrome). Subvalvar PS: surgical resection. Branch PA stenosis: transcatheter balloon dilation and stenting. Emerging: TPVR (Melody, SAPIEN 3, Harmony TPV) for RVOT dysfunction post-repair. MEK inhibitor therapy (trametinib) for Noonan-associated RASopathy PS/HCM (FDA Breakthrough Designation).

		syndromes.	140,000 new/yr Source: Liu et al., IJE 2019; ~8/10,000 births; 8–10% of all CHD		
6	Coarctation of the Aorta	Focal narrowing of the aorta distal to the left subclavian artery. Creates two-zone circulation: HTN upper body, hypotension below.	4-6 per 10,000 live births; ~5-8% of CHD. Males ~2x more affected. US: ~1,440–2,160 new/yr; ~60,000–90,000 pediatric prevalent Global: ~56,000–84,000 new/yr Source: van der Linde et al., JACC 2011; 4–6/10,000 births; males 2x	Echo, CT angiography, or MRI. BP differential between arms and legs is pathognomonic.	SOC: Neonates: surgical resection with end-to-end anastomosis. Older children/adults: transcatheter balloon dilation ± stenting (covered stents preferred). Prostaglandin E1 to maintain ductal patency in critical neonatal coarctation. Emerging: Covered stent technology (Gore Viabahn, NuMED CP stent) for de novo and recurrent coarctation reducing aneurysm risk. Surveillance MRI protocols for long-term aortopathy after repair.
7	Atrioventricular Septal Defect (AVSD) 	Combined atrial + ventricular hole with malformation of the central heart valves. Associated with Down syndrome in 40-50% of cases.	~3 per 10,000 live births; 4-5% of CHD. 60% of CHD in Down syndrome population. US: ~1,080–1,800 new/yr; ~40,000–70,000 pediatric prevalent Global: ~42,000–70,000 new/yr Source: van der Linde et al., JACC 2011; ~3/10,000 births; Down syndrome in ~60% of AVSD cases	Echocardiography. AVSD AI detection app available. Fetal echo for prenatal diagnosis.	SOC: Complete surgical repair (double-patch closure of both ASD and VSD + AV valve reconstruction) in first 3–6 months of life. Pre-operative management of pulmonary hypertension. Emerging: Improved surgical techniques for common AV valve reconstruction; refined timing protocols for partial vs. complete forms. Transcatheter AV valve repair (MitraClip, PASCAL) for recurrent post-operative MR in adults.
8	Partial AVSD (Primum ASD) 	Incomplete form: only the atrial component present - - primum ASD in inferior atrial septum adjacent to AV valves -- without ventricular septal component. AV valves	~1-2 per 10,000 live births; ~30% of all AVSDs. Down syndrome: ~20% (vs ~60% in complete AVSD). US: ~360–720	Echo: primum ASD in inferior septum abutting AV valves; left AV valve cleft on short-axis; assess MR severity. Cardiac MRI. Catheterization for pulmonary pressure if MR	SOC: Surgical repair: primum ASD closure + direct suture closure of left AV valve cleft. Timing: 3–6 months if significant MR; defer to 1–2 years if mild. Operative mortality <1% at experienced centers. Lifelong surveillance; late MR recurrence ~5–10%. Emerging: Transcatheter AV valve repair

		abnormal (left AV valve cleft causing MR) but no inlet VSD. Often misclassified as simple ASD; mitral valve pathology and surgical approach are fundamentally different.	new/yr; ~15,000–30,000 pediatric prevalent Global: ~14,000–28,000 new/yr Source: Partial AVSD ~1–2/10,000 births; ~30% of all AVSDs; Down syndrome in ~20%	severe or delayed presentation.	(MitraClip) for recurrent post-repair MR in adult patients not suitable for reoperation.
9	Tetralogy of Fallot (TOF) 	Four defects: large VSD, overriding aorta, pulmonary stenosis, RV hypertrophy. Most common cyanotic CHD. Virtually all post-repair patients require lifelong follow-up for residual PR, RV dilation, and arrhythmia. Variant: TOF with Absent PV Syndrome (~3-5%) -- massive free PR with aneurysmal PA dilation compressing mainstem bronchi.	3-6 per 10,000 live births; 7-10% of all CHD. Over 350,000 adults with repaired TOF in the US and Europe. US: ~1,080–2,160 new/yr; ~350,000 US total (children + adults with repaired TOF) Global: ~42,000–84,000 new/yr Source: van der Linde et al., JACC 2011; 3–6/10,000 births; 22q11 deletion in ~15%	Echocardiography. Cardiac MRI -- gold standard post-repair: RVEDVi >150-160 mL/m ² = PVR threshold. ECG: QRS >180 ms = independent SCD predictor. Holter. CT for absent PV variant PA anatomy.	SOC: Complete intracardiac repair (VSD closure + RVOT enlargement/transannular patch) in first year. Post-repair PVR when RVEDVi >150 mL/m ² or RVEF <47%. TPVR: Melody (conduit), SAPIEN 3 (conduit/native RVOT), Harmony TPV (native RVOT, FDA 2021). ICD for VT/VF risk. Emerging: Expanded Harmony TPV22 indications for dilated native RVOT. Earlier PVR timing trials to prevent RV dysfunction. TOF with atretic PV: PA reduction plasty + VSD closure + valved conduit. RVOT substrate ablation for refractory VT.
10	D-Transposition of Great Arteries (D-TGA)	Aorta and pulmonary artery are switched -- oxygenated and deoxygenated blood circulate in parallel. Cardiac emergency at birth.	3-5 per 10,000 live births; ~5% of CHD. Male predominance 2-3:1. US: ~1,080–1,800 new/yr; ~70,000–100,000 US total (children + adults) Global: ~42,000–70,000 new/yr Source: van der Linde et al., JACC 2011; 3–5/10,000 births; males 2–3:1	Fetal echocardiography. Pulse oximetry screening at birth. Urgent echo for cyanotic newborn.	SOC: Prostaglandin E1 to maintain PDA. Balloon atrial septostomy (Rashkind procedure, bedside echo guidance) when atrial communication restrictive — life-saving. Arterial switch operation (ASO) within first 2 weeks of life. Emerging: Refined coronary reimplantation techniques for complex coronary anatomy patterns. Long-term neo-aortic root surveillance protocols post-ASO; earlier intervention thresholds for neo-aortic root dilation.
11	Congenital Aortic	Narrowing of the aortic	3-6 per 10,000 live	Echocardiography. Fetal	SOC: Balloon valvuloplasty (acute success

	Stenosis	valve present from birth. Critical neonatal form causes cardiogenic shock when ductus closes.	births; males 3-4x more affected. US: ~1,080–2,160 new/yr; ~60,000–100,000 US total Global: ~42,000–84,000 new/yr Source: van der Linde et al., JACC 2011; 3–6/10,000 births; males 3–4x	echo for prenatal diagnosis. Cardiac catheterization for intervention planning.	>85%) for valvar AS. Ross procedure preferred in children and young adults. TAVR for adolescents/adults with failed prior surgical or Ross procedure. Emerging: Ozaki procedure (aortic valve neocuspidization using autologous pericardium) as Ross alternative — growing evidence. Transcatheter valve-in-valve for failed biological aortic valves. BASILICA technique to prevent coronary occlusion during TAVR.
12	Supravalvar Aortic Stenosis (SVAS) 	Fixed obstruction of ascending aorta at/above sinotubular junction. Two forms: (1) Williams syndrome-associated (~75%) -- ELN deletion 7q11.23, bilateral PA stenosis co-exists; (2) Isolated/non-syndromic (~25%) -- ELN point mutations. Critical: commissures may adhere to supra-aortic ridge, obstructing coronary ostia and risking coronary ischemia -- mandatory intraoperative assessment. Doty three-patch aortoplasty (inverted Y incision, three patch inserts) is technique of choice.	SVAS overall: ~1 in 20,000. Williams syndrome: ~1 in 7,500-10,000; SVAS in ~75%. Isolated: ~1 in 80,000-100,000. Re-stenosis after repair: ~5-10%. US: ~180–360 isolated SVAS cases/yr; ~720–1,080 Williams-related cases/yr Global: ~7,000–14,000 new isolated cases/yr Source: Morris et al., Pediatrics 2010; Williams syndrome 1/7,500–10,000 births; SVAS in ~75% of Williams	Echo: gradient across sinotubular junction; bilateral PA pressures; coronary origins. CT/MRI: full anatomy of ascending aorta, arch, branch PAs, and coronary ostia relative to ridge. Cardiac catheterization: pressure gradient; coronary angiography if coronary ischemia suspected. FISH/CMA for 7q11.23 if Williams suspected.	SOC: Surgical repair indicated for gradient >50 mmHg, symptoms, or LV dysfunction. Doty three-patch aortoplasty preferred. Inspect coronary ostia for ridge adherence at operation. Branch PA stenosis: transcatheter stenting. Lifelong surveillance echocardiography. Emerging: Transcatheter approaches for discrete membranous SVAS. Elastin-targeted therapies for Williams syndrome-related arteriopathy (preclinical). Coronary ostial protection techniques to reduce perioperative SCD risk.
13	Hypoplastic Left Heart Syndrome (HLHS)	Entire left side of heart severely underdeveloped. RV must support both pulmonary and systemic circulation. Most severe CHD.	2-3 per 10,000 live births; 1-2% of CHD but 25-40% of cardiac deaths in week 1 of life. US: ~720–1,080 new/yr; ~20,000–30,000 US total (children + young adults)	Fetal echo (often 18-24 wks). Postnatal: echo, CT/MRI. Assess atrial septum size -- restrictive atrial septum (~5-10% of HLHS) carries significantly higher operative mortality.	SOC: Prostaglandin E1 emergently. Balloon/stent atrial septostomy for restrictive atrial septum. Three-stage Norwood/Glenn/Fontan palliation or Hybrid Norwood. Interstage home pulse oximetry monitoring (reduces interstage mortality from ~15% to ~2–5%). Cardiac transplantation for selected patients. Emerging: Fetal cardiac intervention (balloon aortic valvuloplasty) for borderline

			Global: ~28,000–42,000 new/yr Source: CDC/CHOP; HLHS 2–3/10,000 births; 25–40% of cardiac deaths in week 1 of life		HLHS to promote LV growth. Stem cell therapy (cardiosphere-derived cells — TICAP, PERSEUS trials). Ventricular recruitment strategies. Improved Fontan outcomes with fenestration and thoracic lymphatic interventions.
14	Ebstein's Anomaly	Tricuspid valve leaflets displaced downward into RV, 'atrializing' a portion of it. Associated WPW in 25–30%.	~1 per 20,000 live births; <1% of CHD. Wide severity spectrum. US: ~180–360 new/yr; ~8,000–15,000 US total Global: ~7,000–14,000 new/yr Source: van der Linde et al., JACC 2011; 1/20,000 births; wide severity spectrum	Echocardiography. ECG for WPW. Cardiac MRI for RV function.	SOC: Cone procedure (valve reconstruction using native leaflet tissue) — procedure of choice. ICD for WPW/arrhythmia risk. Catheter ablation for accessory pathway. Bidirectional Glenn shunt for severe RV dysfunction. Emerging: Transcatheter tricuspid valve interventions (CLASP TR, FORMA, TriClip) for patients with residual TR not suitable for reoperation. Refined Cone technique outcomes driving earlier surgical referral.
15	Truncus Arteriosus 🧬	Single great vessel exits the heart instead of two. Complete mixing of oxygenated and deoxygenated blood.	~1 per 10,000 live births; 1–2% of CHD. US: ~360–720 new/yr; ~12,000–20,000 US total Global: ~14,000–28,000 new/yr Source: van der Linde et al., JACC 2011; 1/10,000 births; 22q11 deletion in ~35%	Echocardiography; fetal echo for prenatal diagnosis. Catheterization for pulmonary artery anatomy.	SOC: Surgical repair in neonatal period: VSD closure, separation of pulmonary arteries, RV-to-PA conduit placement. TPVR (Melody/SAPIEN 3/Harmony TPV) for conduit failure — defers or replaces surgical reintervention. Branch PA stenosis: catheter balloon dilation and stenting. Emerging: Tissue-engineered conduits (decellularized homografts, bioresorbable scaffolds) under investigation to reduce reintervention burden. Expanding transcatheter pulmonary valve options for native/dilated RVOT.
16	Interrupted Aortic Arch 🧬	Complete discontinuity of the aortic arch. Survival depends on PDA; cardiovascular collapse when ductus closes. Associated with 22q11 deletion in 50–70%.	~1–2 per 100,000 live births; <1% of CHD. US: ~36–72 new/yr; ~2,000–4,000 US total Global: ~1,400–2,800 new/yr	Fetal/neonatal echocardiography. CT angiography for arch anatomy. Genetic testing for 22q11 deletion.	SOC: Prostaglandin E1 emergently. Surgical arch reconstruction connecting arch segments + simultaneous VSD closure. Emerging: Hybrid and staged approaches for high-risk neonates. Transcatheter management of recurrent arch obstruction post-repair. Genotype-guided counseling for 22q11.2 deletion (~50% of IAA).

			Source: IAA 1–2/100,000 births; 22q11 deletion in ~50–80% type B; <1% of CHD		
17	Pulmonary Vein Disease (TAPVR/PAPVR)	Pulmonary veins drain anomalously -- not into left atrium. Obstructed forms are neonatal emergencies. Post-surgical pulmonary vein stenosis is a distinct, aggressive complication.	1-2 per 10,000 live births; ~2% of CHD. US: ~360–720 new/yr; ~15,000–25,000 US total Global: ~14,000–28,000 new/yr Source: van der Linde et al., JACC 2011; TAPVR 1–2/10,000 births; ~2% of CHD	Echocardiography. CT/MRI for drainage anatomy. Fetal echo for complex forms.	SOC: Urgent surgical repair for obstructed TAPVR (emergency). Elective repair in first weeks/months for unobstructed TAPVR. Sutureless repair technique reduces post-surgical PV stenosis risk. Serial echo surveillance every 3–6 months for ≥2 years post-repair. Emerging: Sutureless 'pericardium-in-situ' technique refinements for reoperations. Systemic antifibrotic therapy (sirolimus, imatinib) for post-operative PV stenosis — active trials. Drug-eluting balloon for transcatheter treatment of stenotic repaired veins.
18	Double Outlet Right Ventricle (DORV)	Both great arteries arise from the RV. Four distinct surgical diagnoses by VSD position: (1) Subaortic VSD -- LV-to-aorta tunnel; (2) Subpulmonary VSD (Taussig-Bing) -- arterial switch; (3) Doubly-committed VSD -- tunnel repair; (4) Remote VSD -- often Fontan palliation. Incorrect anatomic classification leads to the wrong operation.	1-2 per 10,000 live births; ~1-2% of CHD. Taussig-Bing anomaly: ~15% of DORV cases. US: ~360–720 new/yr; ~12,000–20,000 US total Global: ~14,000–28,000 new/yr Source: van der Linde et al., JACC 2011; DORV 1–2/10,000 births; Taussig-Bing ~15% of DORV	Echocardiography. Cardiac MRI -- essential for VSD position, great artery relationships, and coronary anatomy. Catheterization for pulmonary pressure assessment. 3D reconstruction for surgical planning.	SOC: Surgery individualized by VSD location and great artery relationship: intraventricular LV-to-aorta tunnel (REV or Rastelli) for subaortic VSD; arterial switch + VSD closure (Taussig-Bing); staged Fontan for remote VSD or complex anatomy. Emerging: 3D printing and computational flow modeling for pre-operative surgical planning in complex DORV. Hybrid strategies for high-risk neonates. No single DORV repair strategy exists — anatomy determines operation.
19	Double Inlet Left Ventricle (DILV)	Both atria connect to a single dominant left ventricle; RV absent or hypoplastic. Classic single-ventricle anatomy.	~1 per 10,000 live births; ~1% of CHD. US: ~360–720 new/yr; ~10,000–20,000 US total Global: ~14,000–28,000 new/yr Source: DILV	Echocardiography. Cardiac catheterization before Fontan (includes APC mapping and embolization).	SOC: Staged Fontan palliation (bidirectional Glenn → Fontan). Pre-Fontan catheterization: APC embolization (present in ~50–70% at Glenn stage). Heart transplantation for Fontan failure. Emerging: Thoracic lymphatic embolization for Fontan-related PLE and plastic bronchitis. Combined heart-liver

			~1/10,000 births; ~1% of CHD; single ventricular physiology		transplantation for Fontan-associated liver disease (FALD). Mechanical Fontan assist devices under investigation.
20	L-Transposition (L-TGA / ccTGA)	Both AV and VA connections discordant -- 'physiologically corrected' but morphologic RV supports systemic circulation it was not designed for.	~1 per 33,000 live births; <1% of CHD. US: ~110–220 new/yr; ~3,000–6,000 US total Global: ~4,200–8,400 new/yr Source: L-TGA ~1/33,000 births; <1% of CHD; congenitally corrected TGA	Echocardiography. ECG for heart block (develops ~2%/year). Cardiac MRI for RV function tracking.	SOC: Observation if no associated defects and systemic RV function preserved. Double-switch operation (atrial switch + ASO or REV) to restore LV to systemic position. Permanent pacemaker for complete heart block. Emerging: Timing optimization for double-switch based on LV retraining protocols. Transcatheter approaches for associated lesions (VSD device closure, PS balloon dilation). Long-term systemic RV failure surveillance driving earlier double-switch referral.
21	Pulmonary Atresia with Intact VS (PA-IVS)	Atretic (imperforate) pulmonary valve with intact septum; RV develops without normal outflow. RV-dependent coronary circulation in 30-50%.	~1 per 10,000 live births; ~1% of CHD. US: ~360–720 new/yr; ~10,000–18,000 US total Global: ~14,000–28,000 new/yr Source: PA-IVS ~1/10,000 births; ~1% of CHD; spectrum from mild to severe RV hypoplasia	Echocardiography. Cardiac catheterization essential to assess coronary anatomy. Fetal echo.	SOC: Prostaglandin E1. RF perforation + balloon valvuloplasty for adequate RV (biventricular pathway if RV grows). Biventricular or single-ventricle (Fontan) pathway determined by RV size, tricuspid valve annulus z-score, and coronary anatomy (RVDCC). Emerging: Fetal cardiac intervention (in utero pulmonary valvuloplasty) to promote RV growth and potentially avoid univentricular palliation. Refined RV suitability criteria for biventricular vs. one-and-a-half ventricle repair.
22	Pulmonary Atresia with VSD and MAPCAs	Atretic pulmonary valve + hypoplastic/absent pulmonary arteries; lungs supplied by major aortopulmonary collateral arteries (MAPCAs). Catheter-based MAPCA mapping and embolization is a core part of management.	~1-2 per 10,000 live births. US: ~360–720 new/yr; ~8,000–15,000 US total Global: ~14,000–28,000 new/yr Source: PA/VSD/MAPCAs ~1–2/10,000 births; complete repair feasible in ~60–70%	Echocardiography. CT angiography essential for MAPCA mapping. Catheterization for pressure assessment and collateral anatomy.	SOC: Unifocalization surgery (consolidating all MAPCAs into one PA system) + complete intracardiac repair. Multiple staged operations and catheter reinterventions anticipated. Catheter-based MAPCA occlusion for non-unifocalized collaterals causing overcirculation. Emerging: One-stage complete unifocalization + repair (Hanley approach, Stanford) demonstrating improved outcomes over staged repair. Improved conduit options and TPVR to defer surgical reintervention. CT angiography with 3D reconstruction for

					MAPCA mapping.
23	Tricuspid Atresia	Complete absence of tricuspid valve; absent/hypoplastic RV. Left ventricle functions as the sole pump after Fontan palliation.	1-2 per 10,000 live births; 1-3% of CHD. US: ~360–720 new/yr; ~12,000–20,000 US total Global: ~14,000–28,000 new/yr Source: van der Linde et al., JACC 2011; tricuspid atresia 1–2/10,000 births; 1–3% of CHD	Echocardiography. Cardiac catheterization before Fontan (hemodynamics, APC mapping, PA pressure assessment). Fetal echo.	SOC: Staged Fontan palliation (BT shunt → bidirectional Glenn → Fontan). Pre-Fontan catheterization: APC coil embolization, branch PA stenting if needed. Post-Fontan serial catheterizations for APC surveillance and Fontan pathway assessment. Emerging: Fontan optimization with fenestration and thoracic lymphatic embolization for PLE/plastic bronchitis. Mechanical cavopulmonary assist devices (TCPC pump) under investigation. Heart transplantation remains definitive for Fontan failure.
24	Shone's Complex	Four left-sided obstructive lesions: supralvalvar mitral ring, parachute mitral valve, subaortic stenosis, and coarctation of aorta.	Rare; <1 per 100,000 live births. US: ~36–108 new/yr; ~600–1,800 US prevalent Global: ~1,400–4,200 new/yr Source: Shone 1963; <1/100,000 births; 4 left-sided obstructive lesions as defining spectrum	Echocardiography (serial). Cardiac catheterization for hemodynamics. CT/MRI for aortic anatomy.	SOC: Staged surgical relief at each obstructive level (supramitral ring, parachute MV, aortic valve, coarctation). Transcatheter balloon dilation ± stenting of coarctation as bridge or primary treatment. High reintervention rates expected; lifelong surveillance. Emerging: Transcatheter mitral interventions (balloon dilation of supramitral ring, MitraClip for residual MR). Tissue-engineered valves for mitral replacement in growing children (avoiding anticoagulation).
25	Congenital Mitral Valve Anomalies	Spectrum: (1) Parachute Mitral Valve -- all chordae to single papillary muscle; functional MS; part of Shone's complex; (2) Double Orifice Mitral Valve -- two orifices separated by fibrous bridge; (3) Isolated Mitral Leaflet Cleft -- cleft without AVSD; (4) Mitral Atresia -- part of HLHS/single-ventricle spectrum.	Congenital MS: 0.5–1 per 10,000. Isolated mitral cleft: ~0.05% of CHD. Parachute MV: rare, often within Shone's. US: ~180–360 new/yr; ~6,000–12,000 US total Global: ~7,000–14,000 new/yr Source: Congenital MS ~0.5–1/10,000 births; parachute MV <1/100,000;	Echocardiography (2D and 3D TEE essential for subtype distinction and surgical planning). Cardiac MRI for ventricular function. Catheterization for pulmonary pressure. Fetal echo.	SOC: Repair preferred over replacement in children. Isolated mitral cleft: direct suture closure (>90% success). Parachute MV and double-orifice MV: repair tailored to dominant anatomy; replacement often ultimately required. Mitral atresia: Fontan palliation. Emerging: Transcatheter mitral valve repair (MitraClip, PASCAL) for recurrent MR post-surgical repair in adolescents/adults. Tissue-engineered and expandable mitral prostheses to accommodate somatic growth.

			often part of Shone complex		
26	Aortic Arch Abnormalities (Vascular Rings)	Developmental anomalies of aortic arch including double arch, right arch with aberrant subclavian, cervical arch. Vascular rings encircle trachea/esophagus.	1-3% of CHD. Double aortic arch: 40-60% of complete vascular rings. US: ~1,080–3,240 new/yr; ~40,000–100,000 US total Global: ~42,000–126,000 new/yr Source: Vascular rings ~1–3% of CHD; double aortic arch 40–60% of complete rings	Echocardiography. CT angiography gold standard for ring anatomy. Bronchoscopy for tracheomalacia.	SOC: Surgical ring division (ligating minor arch/ligamentum arteriosum). Excellent outcomes at experienced centers. Tracheobronchomalacia managed with airway stenting or aortopexy if persistent post-ring division. Emerging: Thoracoscopic (minimally invasive) ring division — comparable outcomes to open surgery with faster recovery. Pre-operative 3D CT reconstruction and virtual surgical planning for complex double aortic arch.
27	ALCAPA	Left coronary artery arises from the pulmonary artery rather than the aorta, causing myocardial ischemia from birth. Presents with infantile heart failure or sudden death.	~1 in 300,000 live births. Accounts for ~5% of infant cardiac deaths if unrepaired. US: ~12–36 new/yr; ~500–1,000 US total Global: ~470–940 new/yr Source: ALCAPA ~1/300,000 births; accounts for ~5% of infant cardiac deaths if unrepaired	Echocardiography (retrograde coronary flow on Doppler). CT coronary angiography. ECG may show anterolateral ischemia.	SOC: Surgical coronary reimplantation (transfer to aorta) — curative if performed early. ECMO for hemodynamic instability. LV function often recovers substantially with unloading and revascularization. Emerging: Improved post-operative LV recovery monitoring with cardiac MRI. Ventricular assist devices as bridge to recovery. Myocardial viability MRI to guide expectation of LV functional recovery.
28	AAOCA (Anomalous Aortic Origin of Coronary Artery)	Coronary artery arises from the wrong sinus or follows an interarterial course between aorta and pulmonary artery. Risk of sudden cardiac death, especially with exertion.	~0.1-0.3% of general population. AAOCA of left coronary is highest risk for SCD. US: ~3,600–10,800 US total all ages; ~1,000–3,000 pediatric Global: ~100,000–300,000 global (all ages)	Echocardiography (limited). CT coronary angiography -- gold standard. Exercise stress testing for ischemia.	SOC: Surgical unroofing of interarterial segment (preferred for high-risk anatomy). Exercise restriction pre-operatively. Coronary reimplantation or bypass in select cases. Emerging: Risk stratification protocols (FFR-CT, stress perfusion MRI, exercise stress echo) to refine surgical indications and identify truly flow-limiting lesions. Watchful waiting evidence expanding for low-risk anatomy.

			Source: Davis et al., JACC 2016; AAOCA ~0.1–0.3% of general population		
29	Heterotaxy Syndrome / Atrial Isomerism	Abnormal left-right organ arrangement combined with complex intracardiac defects. Right isomerism (asplenia) associated with severe cyanotic CHD; left isomerism (polysplenia) with heart block and interrupted IVC.	~1-2 per 10,000 live births; ~3% of all CHD. Right isomerism: ~50-70% mortality in first year without intervention. US: ~360–720 new/yr; ~10,000–18,000 US total Global: ~14,000–28,000 new/yr Source: Heterotaxy ~1–2/10,000 births; ~3% of CHD; right isomerism ~50–70% mortality without intervention	Fetal/neonatal echocardiography. CT/MRI for cardiac and visceral situs. Abdominal ultrasound for spleen. Genetic testing. ECG for heart block.	SOC: Highly individualized: most require staged Fontan palliation. Asplenia: lifelong penicillin prophylaxis ± pneumococcal/meningococcal vaccines. Pacemaker for complete heart block (left isomerism). Emerging: Improved AV valve repair strategies for complex heterotaxy valve anatomy. Bilateral SVC management optimization. Liver laterality assessment for equitable hepatic flow distribution in Fontan. Lymphatic interventions for post-Fontan complications.
30	Aortopulmonary Window	Rare communication between the ascending aorta and pulmonary artery, distinct from PDA. Causes massive left-to-right shunting and neonatal heart failure.	~1 per 100,000 live births; <1% of CHD. US: ~36–72 new/yr; ~1,000–2,000 US total Global: ~1,400–2,800 new/yr Source: Aortopulmonary window ~1/100,000 births; <1% of CHD	Echocardiography. CT angiography for anatomy. Cardiac catheterization for pulmonary vascular resistance.	SOC: Surgical repair (direct closure or patch) in neonatal period. Transcatheter device closure for suitable anatomy (distant from coronary ostia and pulmonary valves). Emerging: Expanded catheter-based closure with Amplatzer Duct Occluder II and custom devices for AP windows with suitable anatomy.
31	Cor Triatriatum	A fibromuscular membrane divides the left atrium into two chambers, creating pulmonary venous obstruction mimicking mitral stenosis.	Rare; ~0.1-0.4% of CHD. Often presents in infancy with pulmonary edema. US: ~36–144 new/yr; ~1,500–4,000 US total Global: ~1,400–	Echocardiography (TTE/TEE: membrane in left atrium with gradient). Cardiac MRI for complex anatomy.	SOC: Surgical membrane resection — curative in nearly all cases. Excellent outcomes with early repair before pulmonary hypertension develops. Emerging: Transcatheter balloon membrane dilation or perforation as bridge to surgery in high-risk neonates with obstructed cor triatriatum (very rare). Robot-assisted minimally invasive resection at specialized

			5,600 new/yr Source: Cor triatriatum ~0.1–0.4% of CHD; fibromuscular membrane obstructing left atrium		centers.
32	Discrete Subaortic Stenosis	Fibromuscular membrane or ridge immediately below the aortic valve in the LVOT causing fixed subvalvar obstruction. Progressive obstruction damages the aortic valve leaflets from the turbulent jet (causing AR). Recurrence after surgical resection ~10-20%. Strong male predominance.	~1 in 30,000 live births; ~15-20% of all congenital LVOT obstruction. May appear first in childhood or adolescence. Males 2-3x more affected. US: ~120–240 new/yr; ~5,000–10,000 US total Global: ~4,700–9,400 new/yr Source: DSS ~1/30,000 births; ~15–20% of all LVOT obstruction; fibromuscular ridge may form post-birth	Echo: subvalvar membrane with peak LVOT gradient; assess aortic valve for secondary AR. Cardiac MRI: LVOT anatomy. Catheterization: LVOT pressure pullback. TEE for precise membrane localization.	SOC: Surgical membrane resection ± septal myectomy for gradient >30–40 mmHg or any developing AR. Balloon dilation ineffective. Post-op surveillance every 1–2 years lifelong; 10–20% recurrence rate. AV repair/replacement if secondary AR severe. Emerging: Earlier surgical intervention protocols to prevent progressive AR — intervention before AR becomes significant. Cardiac MRI tissue characterization to distinguish fibrous vs. muscular/tunnel LVOT obstruction for surgical planning.
33	Sinus of Valsalva Aneurysm	Congenital aneurysm of an aortic sinus (right coronary ~70%; non-coronary ~25%). Rupture causes acute left-to-right shunt, continuous murmur, rapid heart failure, and hemodynamic collapse. Right coronary sinus aneurysm ruptures most commonly into the RV. Associated with VSD in ~30-50% of cases.	~0.09% of CHD; ~1 in 100,000. Predominantly diagnosed in adulthood when rupture occurs. Males 4x more affected. Associated VSD in ~30-50%. US: ~36–72 new/yr; ~2,000–5,000 US total (predominantly adult-diagnosed) Global: ~1,400–2,800 new/yr Source: Sinus of Valsalva aneurysm	Echo (TTE/TEE): aneurysmal sac; Doppler continuous high-velocity shunt on rupture. CT angiography: gold standard for anatomy. Cardiac catheterization: Qp/Qs, coronary anatomy.	SOC: Surgical patch closure from both aortic root and receiving cardiac chamber simultaneously; associated VSD closed at same operation. Emergency surgery for acute rupture with hemodynamic instability. Emerging: Transcatheter device closure (Amplatzer Duct Occluder, Vascular Plug) for suitable non-ruptured anatomy. Endovascular stent-graft for selected high-surgical-risk patients.

			~0.09% of CHD; ~1/100,000; predominantly right coronary sinus		
34	Scimitar Syndrome (Pulmonary Venolobar Syndrome)	Rare anomaly: entire right lung drains via large venous channel (the 'scimitar vein') into IVC rather than left atrium. Also includes: right lung hypoplasia and sequestration, dextroposition of heart, and in ~30% an aberrant systemic arterial supply to the right lower lobe. Distinct from TAPVR: involves only right lung, includes pulmonary hypoplasia, requires different surgical approach.	~1-3 per 100,000 live births. Female predominance 2:1. Neonatal presentation (severe form with right lung hypoplasia and pulmonary hypertension): ~25% of cases -- high mortality without intervention. US: ~36–108 new/yr; ~1,200–3,000 US total Global: ~1,400–4,200 new/yr Source: Scimitar syndrome ~1–3/100,000 births; female predominance 2:1	Chest X-ray: pathognomonic 'scimitar sign' -- curvilinear density paralleling right heart border. CT angiography: gold standard -- scimitar vein course, drainage site, anomalous aortic supply, right lung hypoplasia. Echo. Cardiac catheterization: PVR, selective aortography for anomalous arterial supply.	SOC: Surgical redirection of scimitar vein to left atrium via intra-atrial baffle or direct reimplantation; concurrent ligation of aberrant aortic arteries. Observation for mild asymptomatic adult forms (Qp/Qs <1.5). Emerging: Transcatheter baffle creation with stent-based redirection of scimitar vein (avoids cardiopulmonary bypass in selected cases). Combined embolization of aberrant aortic collaterals + transcatheter venous redirection.
35	Unroofed Coronary Sinus / PLSVC Draining to Left Atrium	Congenital absence of roof of coronary sinus creating communication with left atrium. Underlying anatomy: PLSVC drains into coronary sinus which then drains into LA, causing right-to-left shunt and cyanosis. Classic missed diagnosis at ASD repair: patching the ASD without recognizing unroofed coronary sinus causes persistent post-operative cyanosis. PLSVC itself (~3-10% of CHD) is usually benign but must be identified before	Unroofed coronary sinus: ~1% of all CHD. PLSVC: ~3-10% of CHD overall; ~0.3% of general population. Complete unroofing: ~0.1% of CHD. US: ~720–3,600 new cases/yr (PLSVC + unroofed CS combined); ~25,000–75,000 US total Global: ~28,000–140,000 new/yr Source: PLSVC ~3–10% of CHD;	Echo: dilated coronary sinus; bubble contrast study -- microbubbles appear in LA from left arm injection (diagnostic for PLSVC-to-LA drainage). CT angiography: gold standard for PLSVC anatomy and coronary sinus continuity. Cardiac catheterization: O2 saturation step-up/step-down; selective venography.	SOC: Surgical repair: (1) PLSVC ligation if right SVC present and bridging vein adequate + coronary sinus closure; (2) Baffling to redirect flow to RA; (3) PLSVC reimplantation to RA appendage if no right SVC. Pre-operative identification mandatory before any cardiac surgery. Emerging: Transcatheter occlusion of PLSVC in selected adult patients with adequate collateral drainage. Identification protocols (routine pre-op echo screening) to prevent inadvertent PLSVC ligation.

		any cardiac operation.	unroofed coronary sinus ~1% of CHD; often clinically silent		
CARDIOMYOPATHIES					
#	CONDITION	DESCRIPTION	FREQUENCY	DIAGNOSIS	TREATMENT
36	Hypertrophic Cardiomyopathy (HCM) 🧬	Abnormal LV wall thickening from sarcomere gene mutations. Obstructive form (HOCM) causes outflow obstruction. Leading cause of sudden death in young athletes.	~1 in 500 individuals (~700,000 Americans). Leading cause of SCD in athletes under 35 (~36% of events). US: ~1,460,000 US total all ages; ~200,000–300,000 US pediatric/adolescent Global: ~50,000,000–60,000,000 global (all ages) Source: Maron et al., JACC 2022; HCM ~1/500 individuals; ~700,000 Americans	Echocardiography (septal thickness ≥ 15 mm or ≥ 13 mm with family history). Genetic testing. CMR for fibrosis (LGE). ECG screening.	SOC: Beta blockers, calcium channel blockers for symptom management. Septal myectomy (surgical gold standard for refractory LVOTO). Alcohol septal ablation for appropriate anatomy. ICD for high-risk patients (SCD prevention). Emerging: Mavacamten (Camzyos, FDA 2022) — cardiac myosin inhibitor; first disease-modifying drug for obstructive HCM. Aficamten (Myqorzo, FDA 2025) — next-generation myosin inhibitor. Expanding use in pediatric/adolescent populations.
37	Dilated Cardiomyopathy (DCM) 🧬	LV (and often both ventricles) dilated with impaired contractility. Most common cardiomyopathy requiring pediatric transplantation.	0.57-1.13 per 100,000 children/year. Up to 50-60% idiopathic; 20-35% genetic. US: ~200–400 new pediatric cases/yr; ~5,000–10,000 US pediatric prevalent Global: ~7,700–15,400 new pediatric cases/yr Source: Towbin et al., Circulation 2006; pediatric DCM 0.57–1.13/100,000	Echocardiography (LVEF $< 55\%$, LV dilation). Cardiac MRI. Genetic panel. Endomyocardial biopsy if myocarditis suspected.	SOC: ACE inhibitors/ARBs, beta blockers, diuretics, aldosterone antagonists. Sacubitril/valsartan (Entresto) for pediatric HF (FDA-approved ≥ 1 year). CRT for dyssynchrony. ICD. VAD as bridge to transplantation (Berlin Heart EXCOR < 15 kg; continuous-flow for larger). Cardiac transplantation. Emerging: SGLT2 inhibitors (dapagliflozin, empagliflozin) expanding to pediatric HF with emerging evidence. Gene therapy for genetic DCM subtypes (LMNA, TTN, RBM20) in early trials. Precision medicine genotype-guided therapy.

			children/yr		
38	Restrictive Cardiomyopathy (RCM)	Stiff, non-compliant ventricles with elevated filling pressures. Systolic function initially preserved. Rarest and worst-prognosis pediatric cardiomyopathy.	0.03-0.04 per 100,000 children/year; 2-5% of pediatric cardiomyopathies. US: ~22–29 new pediatric cases/yr; ~500–1,000 US pediatric prevalent Global: ~840–1,120 new pediatric cases/yr Source: Colan et al., JACC 2007; RCM 0.03–0.04/100,000 children/yr; 2–5% of pediatric CMs	Echocardiography (biatrial enlargement, normal LV size/EF, restrictive filling pattern). Cardiac MRI. Cardiac catheterization.	SOC: Diuretics for symptomatic relief only. Cardiac transplantation is the only effective long-term treatment; early listing is standard practice. Emerging: mTOR pathway modulation for sarcomeric RCM. Gene therapy targeting MYH7, TNNI3 variants under investigation. Refined transplant listing criteria to optimize timing.
39	ARVC (Arrhythmogenic Right Ventricular Cardiomyopathy) 🧬	Genetically defined (desmosomal gene mutations: PKP2, DSP, DSG2) cardiomyopathy with fibro-fatty replacement of RV myocardium. Causes ventricular arrhythmias triggered by exercise; leading cause of SCD in young athletes.	~1 in 2,000-5,000. Accounts for ~11-17% of SCD in athletes under 35. PKP2 mutations in ~40% of familial cases. US: ~14,600–36,500 US total all ages; ~3,000–7,000 US pediatric Global: ~580,000–1,460,000 global (all ages) Source: Corrado et al., Circulation 2017; ARVC ~1/2,000–5,000; PKP2 most common mutation	Task Force Criteria (revised 2020). Cardiac MRI (fibro-fatty infiltration, RV dilation, RVEF). ECG: epsilon waves, T-wave inversions V1-V3. Holter. Exercise stress test. Genetic panel (desmosomal genes). Endomyocardial biopsy.	SOC: Exercise restriction — critical and mandatory. ICD for sustained VT, VF, or syncope. Catheter ablation for refractory VT (epicardial approach often required). Beta blockers. Heart transplantation for advanced biventricular failure. Emerging: Flecainide as add-on antiarrhythmic therapy. Gene-silencing approaches (ASO, siRNA) for PKP2 pathogenic variants in preclinical development. Sports cardiology risk stratification refinements for risk-stratified return-to-play decisions.
40	Left Ventricular Non-Compaction (LVNC) 🧬	Distinct cardiomyopathy characterized by prominent trabeculations and a non-compacted inner myocardial layer. Associated with multiple	Estimated 0.05-0.24% in echocardiographic studies. Present in ~10% of pediatric cardiomyopathies.	Echo: Jenni criteria (NC/C ratio >2 in adults, >1.4 in children in systole). Cardiac MRI. Genetic panel. ECG (LBBB, WPW pattern).	SOC: Standard heart failure therapy (ACEi/ARB, beta blocker, diuretics). Anticoagulation for LVEF <35% or thrombus. ICD for high-risk arrhythmia. Cardiac transplantation for refractory HF. Emerging: Genetic phenotype-specific

		genetic mutations (MYH7, MYBPC3, LMNA, TAZ). Causes heart failure, ventricular arrhythmias, and thromboembolic risk.	US: ~36,500–175,200 US total (echocardiographic); clinical LVNC ~10,000–30,000 Global: ~1,150,000–5,520,000 global (echocardiographic); clinical significantly lower Source: Towbin et al., JACC 2022; echocardiographic prevalence 0.05–0.24%; significant diagnostic variability between Jenni and Petersen criteria		management — reclassification of pathological vs. physiological non-compaction using refined MRI criteria (NCC/CC ratio, strain imaging). Emerging evidence that isolated LVNC may not require therapy without LV dysfunction.
41	Duchenne MD Cardiomyopathy 	Progressive dilated cardiomyopathy from dystrophin gene mutations. Cardiac involvement is nearly universal by age 18; now the leading cause of DMD death.	1 in 3,500-5,000 male births. Cardiomyopathy present in virtually all patients by age 18. US: ~1,000–1,400 new US male cases/yr; ~75,000–100,000 US Duchenne total Global: ~40,000–56,000 new global male cases/yr Source: Hoffman et al., NEJM 2019; DMD 1/3,500–5,000 male births; cardiomyopathy present virtually all by age 18	Annual echocardiography + ECG from age 10 (or at diagnosis). Cardiac MRI with LGE for early fibrosis detection. LVEF monitoring.	SOC: ACE inhibitors (cardioprotective even pre-dysfunction — start by age 10). Beta blockers once cardiac dysfunction present. Corticosteroids (deflazacort preferred). Ventilatory support. ICD consideration for arrhythmia. Emerging: Delandistrogene moxeparvovec (Elevidys, Sarepta, FDA 2023/2024) — micro-dystrophin gene therapy for ambulatory patients ≥4 years. Exon-skipping therapies (eteplirsen, golodirsen, viltolarsen, casimersen) for applicable mutations. Givinostat (HDAC inhibitor, FDA 2024) for muscle and cardiac preservation.
42	Emery-Dreifuss MD / LMNA Cardiomyopathy 	Caused by LMNA or EMD gene mutations. Classic triad: early joint contractures, slowly progressive muscular	EDMD overall: ~1 in 100,000. SCD risk in LMNA cardiac disease: >5%/year in high-risk	ECG: atrial standstill, progressive heart block (pathognomonic for EDMD). Echocardiography (DCM).	SOC: Early and aggressive ICD implantation — mainstay even before significant DCM (LMNA mutations carry very high sudden death risk). Pacemaker for symptomatic bradycardia/heart block. Standard HF

		dystrophy, and life-threatening cardiac conduction disease progressing to DCM. Highest SCD risk of any inherited condition.	subgroups. US: ~36–72 new/yr; ~600–1,200 US total Global: ~1,400–2,800 new/yr Source: Emery-Dreifuss MD ~1/100,000; LMNA cardiac disease SCD risk >5%/yr without ICD	Holter for conduction disease and VT. Genetic testing (LMNA, EMD).	therapy. Cardiac transplantation for advanced disease. Emerging: Temsirolimus (mTOR inhibitor) in early trials for LMNA-DCM. SGLT2 inhibitors for laminopathy-associated HF. Gene therapy for LMNA haploinsufficiency in preclinical stage.
43	Myocarditis	Inflammation of heart muscle, most commonly viral (enterovirus, adenovirus, parvovirus B19). Ranges from mild to fulminant cardiogenic shock.	~1-2 per 100,000 children/year. Significantly underdiagnosed. Fulminant form in ~15-20% of pediatric cases. US: ~200–400 new US pediatric clinical cases/yr (true incidence ~5× higher) Global: ~7,700–15,400 new pediatric cases/yr Source: Freedman et al., Circulation 2016; myocarditis ~1–2/100,000 children/yr; significantly underdiagnosed	Echo (LV dysfunction). Cardiac MRI (gold standard: T2 edema, LGE fibrosis, Lake Louise criteria). Endomyocardial biopsy for definitive diagnosis.	SOC: Supportive care. MCS: ECMO as rescue for fulminant cardiogenic shock; VAD (Berlin Heart EXCOR <15 kg or continuous-flow for larger) as bridge to recovery or transplantation. Immunosuppression for giant cell/eosinophilic/autoimmune myocarditis. Heart transplantation. Emerging: Endomyocardial biopsy-guided and cardiac MRI-guided immunosuppression protocols (corticosteroids + azathioprine for lymphocytic myocarditis per updated European guidelines). Anakinra (IL-1 antagonist) for immune-mediated myocarditis with hemodynamic compromise. Pembrolizumab discontinuation + corticosteroids for immune checkpoint inhibitor myocarditis.
44	Pompe Disease (Glycogen Storage Disease Type II) 🧬🧬🧬	Autosomal recessive lysosomal acid alpha-glucosidase (GAA) deficiency. Classic infantile form: massive hypertrophic cardiomyopathy + hypotonia + respiratory failure within first year.	Classic infantile Pompe: ~1 in 138,000 live births. All forms combined: ~1 in 40,000. US: ~26–52 new US infantile cases/yr; ~90 new US all-forms cases/yr Global: ~1,000–	Newborn screening (GAA enzyme activity). GAA enzyme assay in leukocytes. GAA gene sequencing. Echo (massive HCM). ECG (giant QRS complexes, shortened PR interval). CK elevation.	SOC: ERT: alglucosidase alfa (Myozyme/Lumizyme, Sanofi) — initiate as early as possible; cardiac function often normalizes with early treatment. Respiratory support. Newborn screening allows pre-symptomatic treatment. Emerging: Next-generation ERT: cipaglucosidase alfa + miglustat (Pombiliti + Opfolda, Amicus, FDA 2023) — improved glycan-independent cellular uptake, superior

			2,000 new infantile cases/yr globally Source: van Capelle et al., Orphanet 2016; classic infantile Pompe ~1/138,000 births; all forms ~1/40,000		to alglucosidase alfa in late-onset disease. Gene therapy: AAV-based approaches (AT845, Spark Therapeutics) in Phase 1/2 trials. mRNA replacement therapy under investigation.
45	Danon Disease 	X-linked lysosomal storage disorder (LAMP2 gene mutations). Classic triad: HCM, skeletal myopathy, and intellectual disability. WPW syndrome in ~70% of males. Rapidly progressive HCM often requiring transplantation.	~1 in 40,000. Estimated ~1% of HCM referrals. Frequently misdiagnosed as sarcomeric HCM. US: ~180–360 US total all ages; ~50–100 US pediatric cases Global: ~7,000–14,000 global (all ages) Source: Danon disease ~1/40,000; X-linked LAMP2 mutation; ~1% of HCM referrals	Echo (severe HCM, often massive). ECG: WPW pattern, extremely high voltage. Lysosomal enzyme panel (LAMP2 protein absent on immunostaining). Genetic testing (LAMP2). Muscle biopsy. Cardiac MRI (LGE fibrosis).	SOC: ICD for SCD risk (combination of WPW + severe HCM). Catheter ablation for WPW accessory pathway. Cardiac transplantation — often the only effective long-term therapy; median age at transplantation ~20 years in males. Emerging: Gene therapy: AT845 (Astellas, AAV9-LAMP2B) in Phase 1/2 trials — promising early results for LAMP2 protein restoration. Female carriers require close cardiac surveillance (later-onset, variable penetrance).
46	Fabry Disease (Cardiac) 	Lysosomal storage disorder (alpha-galactosidase A deficiency). Cardiac manifestations: HCM, arrhythmia, heart failure.	~1 in 40,000 (general); possibly 1 in 3,100 including late-onset variants. X-linked recessive. US: ~1,800–8,700 US total (including late-onset variants); ~500–1,000 US pediatric Global: ~70,000–340,000 global (all ages, including late-onset variants) Source: Mehta et al., Genet Med 2020; classical Fabry ~1/40,000;	Enzyme activity (alpha-gal A) in leukocytes. Genetic testing. Echo and cardiac MRI. Plasma/urine Gb3. Newborn screening in some states.	SOC: ERT: agalsidase alfa (Replagal, EU) or agalsidase beta (Fabrazyme, Sanofi, US). Treat cardiac complications: beta blockers/CCBs for HCM, anticoagulation for stroke risk, ICD/pacemaker for arrhythmia and heart block. Emerging: Migalastat (Galafold, Amicus) — oral pharmacological chaperone for patients with amenable GLA mutations (~35–50% of patients); non-inferior to ERT. Next-generation ERT: pegunigalsidase alfa (PRX-102, Chiesi, EU approved) for superior stability and reduced immunogenicity. Gene therapy trials: ST-920 (4D Molecular Therapeutics), 4D-310 in Phase 1/2.

			NBS studies suggest ~1/3,100 including late-onset variants		
47	Friedreich's Ataxia (Cardiac) 🧬🧬🧬	GAA repeat expansion in frataxin gene causes mitochondrial iron accumulation. Cardiac involvement (HCM + arrhythmia) present in >90% and is the leading cause of death.	~1 in 50,000; most common inherited ataxia. US: ~730–1,460 US total with cardiac involvement; ~500–1,000 US pediatric Global: ~29,000–58,000 global Source: Durr, Lancet Neurol 2010; Friedreich's ataxia ~1/50,000; HCM in ~80% of patients	Echocardiography (LV hypertrophy). Cardiac MRI. ECG (T-wave inversion common). Genetic testing (GAA repeat length).	SOC: Beta blockers for symptomatic HCM. ACE inhibitors for heart failure. ICD consideration for malignant arrhythmia/sustained VT. Cardiac surveillance annual echo + ECG. Emerging: Omaveloxolone (Skyclarys, FDA 2023) — first disease-modifying therapy for FA (age ≥16); reduces neurological progression; cardiac effects under prospective evaluation. Frataxin replacement: mRNA therapy (Larimar, CTI-1601) in Phase 2; AAV-FXN gene therapy in early Phase 1. Epigenetic reactivation of silenced frataxin allele (HDAC inhibitors).
48	Barth Syndrome (Cardiac) 🧬🧬🧬	X-linked disorder (tafazzin gene mutations) with abnormal cardiolipin metabolism. Presents with DCM, skeletal myopathy, neutropenia, and growth retardation.	~1 in 300,000-400,000 male births. Significantly underdiagnosed. US: ~9–12 new US male cases/yr; ~200–500 US total Global: ~350–470 new global male cases/yr Source: Barth Syndrome Foundation; ~1/300,000–400,000 male births; X-linked TAZ mutation; significantly underdiagnosed	Echocardiography (DCM; occasionally non-compaction). Urine 3-methylglutaconic acid. Plasma/urine monolysocardiolipin ratio. Genetic testing (TAZ gene).	SOC: Standard HF therapy (ACE inhibitors, beta blockers, diuretics). Nutritional support (leucine supplementation). ICD for ventricular arrhythmia. Cardiac transplantation for refractory HF. Emerging: Elamipretide (SS-31, mitochondria-targeted peptide, Stealth BioTherapeutics) — Phase 2 trial in Barth syndrome (TAZPOWER trial). TAZ gene therapy (AAV-mediated tafazzin restoration) in preclinical stage.
49	Mucopolysaccharidoses (Cardiac) 🧬🧬🧬	Lysosomal storage disorders (MPS I/Hurler, MPS II/Hunter, MPS IVA/Morquio, MPS VI) causing progressive glycosaminoglycan	Combined MPS incidence: ~1 in 25,000-30,000 live births. Cardiac involvement present in >90%.	Echocardiography (serial: valve thickening, regurgitation/stenosis progression). Urine glycosaminoglycans. Leukocyte enzyme activity	SOC: ERT: laronidase (MPS I/Aldurazyme), idursulfase (MPS II/Elaprase), galsulfase (MPS VI/Naglazyme), elosulfase alfa (MPS IVA/Vimizim). HSCT for MPS I Hurler before age 2.5 years (prevents cognitive decline, cardiac benefit partial). Surgical valve

		accumulation in cardiac valves, coronary arteries, and myocardium.	US: ~120–144 new US cases/yr (all MPS combined); ~10,000–20,000 US total Global: ~4,700–5,600 new global cases/yr Source: Khan et al., Orphanet 2017; combined MPS ~1/25,000–30,000 births; cardiac valvular involvement in virtually all MPS types	assay. Genetic testing.	repair/replacement for advanced valve disease. Emerging: Intrathecal/intracerebroventricular ERT to address CNS disease (not reached by IV ERT). Gene therapy: in vivo AAV and ex vivo HSCT-gene therapy combinations for MPS I, II, III (active Phase 1/2 trials). Substrate reduction therapy (SRT) with genistein, miglustat. Genome editing (ZFN, CRISPR) preclinical.
50	Mitochondrial Cardiomyopathy 	Cardiomyopathy (most commonly HCM, less commonly DCM) caused by mutations in mitochondrial DNA or nuclear genes encoding mitochondrial proteins -- failure of oxidative phosphorylation in energy-demanding cardiomyocytes. Major syndromes: MELAS, Kearns-Sayre (complete heart block + HCM), Leigh syndrome (severe infantile HCM), MERRF. Unlike sarcomeric HCM, myectomy and septal reduction are rarely appropriate. Cardiac transplantation is controversial because multisystem disease continues after transplant.	Mitochondrial disease overall: ~1 in 5,000-10,000. Cardiac involvement in ~20-40%. Kearns-Sayre: ~1 in 100,000; complete heart block in virtually all. MELAS: ~1 in 15,000-25,000. US: ~730–1,460 new US cases/yr; ~12,000–24,000 US total with cardiac involvement Global: ~28,000–56,000 new global cases/yr Source: Mitochondrial disease ~1/5,000–10,000; cardiac involvement in ~20–40% of cases	Echo: HCM pattern most common. ECG: complete heart block, pre-excitation, or conduction defects (especially KSS). Serum and CSF lactate (elevated). Muscle biopsy: ragged red fibers. Mitochondrial DNA sequencing; nuclear gene panel. Brain MRI (Leigh: symmetric basal ganglia T2 hyperintensities).	SOC: No disease-modifying therapy proven to alter cardiac course. Standard HF therapy. Mandatory pacemaker for complete heart block (KSS — sudden death is a leading cause). ICD for sustained VT/VF. CoQ10 supplementation (widely used). Avoid valproate (complex I inhibitor). Emerging: NAD ⁺ precursors (nicotinamide riboside, NMN) under investigation. Vatiquinone (EPI-743) in pediatric mitochondrial disease trials. Idebenone for LHON-related mitochondrial disease. Mitochondrial replacement therapy (MRT, 'three-parent IVF') for maternal inheritance prevention — approved in UK, in use clinically. Cardiac transplantation after careful multidisciplinary review; multisystem disease progression continues post-transplant.
CONNECTIVE TISSUE DISORDERS / AORTOPATHIES					
#	CONDITION	DESCRIPTION	FREQUENCY	DIAGNOSIS	TREATMENT

51	Marfan Syndrome (Cardiac) 🧬🧬🧬	Autosomal dominant connective tissue disorder (FBN1 gene). Cardinal cardiac features: progressive aortic root dilation (risk of dissection and rupture), aortic valve regurgitation, and mitral valve prolapse.	~1 in 5,000 individuals (~65,000 Americans). Aortic dissection risk: ~5-10%/year once root >5cm. US: ~13,000–15,000 US total; ~3,000–4,000 US pediatric Global: ~520,000–600,000 global Source: De Paepe et al., Am J Med Genet 1996; Marfan ~1/5,000; ~65,000 Americans confirmed	Ghent criteria 2010. Echocardiography (aortic root Z-score). Annual cardiac MRI/CT. Slit-lamp exam (ectopia lentis). Genetic testing (FBN1). Family screening.	SOC: Losartan or atenolol — reduce aortic root growth rate. Prophylactic aortic root replacement when root >4.5–5.0 cm (or aortic size index >2.75 cm/m ²). Valve-sparing root repair preferred in young patients. Exercise restriction. Annual echocardiographic surveillance. Emerging: TGF-β pathway inhibitors beyond losartan under investigation. Combined ARNI (sacubitril/valsartan) for Marfan-associated LV dysfunction. FBN1 genotype-specific surgical thresholds; aortic size index and growth rate algorithms replacing absolute diameter alone.
52	Loeys-Dietz Syndrome 🧬🧬🧬	Autosomal dominant aortopathy (TGFB1, TGFB2, SMAD3, TGFB2/3 mutations). More aggressive aortic dilation and dissection at smaller diameters than Marfan syndrome.	~1 in 50,000–100,000. TGFB1/2 mutations carry highest risk. US: ~730–1,460 US total; ~200–400 US pediatric Global: ~29,000–58,000 global Source: Loeys-Dietz syndrome ~1/50,000–100,000; TGFB1/2, SMAD2/3, TGFB2/3 mutations	Genetic testing (TGFB1/2, SMAD3, TGFB2/3 panel). Echocardiography. Full aortic MRI/CT from head to pelvis -- LDS affects entire arterial tree. Annual surveillance imaging.	SOC: Losartan — first-line therapy. Lower threshold for prophylactic aortic surgery (~4.0–4.5 cm vs. 5.0 cm in Marfan). Pan-arterial surveillance (head-to-pelvis MRA annually). Exercise restriction. Emerging: Genotype-specific thresholds: TGFB1 vs. TGFB2 vs. SMAD3 vs. TGFB2 variants have differing natural histories. Targeted TGFB pathway inhibitors under investigation. Cervical artery surveillance protocols standardizing.
53	Turner Syndrome (Cardiac) 🧬🧬🧬	Sex chromosome aneuploidy (45,X or mosaic). Cardiac findings in ~30-50%: bicuspid aortic valve (~16%), coarctation (~10%), aortic dilation. Dissection risk 6–10x higher than general population despite relatively modest aortic dilation.	~1 in 2,000–2,500 live female births. Cardiovascular disease is the leading cause of premature mortality. US: ~29,000–36,500 new US cases/yr (all Turner); ~12,000–18,000 cardiac	Echocardiography at diagnosis (BAV, coarctation). MRI/CT for full aortic assessment. BP in all four extremities. Karyotype. Aortic size index (ASI) -- dissection risk increases significantly at ASI >2.5 cm/m ² .	SOC: Estrogen replacement therapy (starting ~12 years for puberty induction). Antihypertensive therapy. Prophylactic aortic surgery at lower thresholds than general population (ASI >2.5 cm/m ²). Coarctation repair per standard protocols. Lifelong cardiology surveillance. Emerging: Refined aortic size index thresholds incorporating aortic area/height index. Growth hormone effects on aortic dimensions under study. Reproductive

			cases/yr Global: ~1,120,000–1,400,000 new global cases/yr (all Turner) Source: Turner syndrome ~1/2,000–2,500 live female births; BAV in ~30–50%; CoA in ~15%		planning with high-risk obstetric centers for those attempting pregnancy.
54	Vascular Ehlers-Danlos Syndrome (EDS Type IV) 🧬🧬🧬	Autosomal dominant (COL3A1 mutations encoding type III procollagen). Most life-threatening EDS subtype. Unlike Marfan/Loeys-Dietz, the aortic root is often normal -- danger lies in spontaneous rupture of medium-sized arteries WITHOUT preceding aneurysm. Arterial tissue is extraordinarily fragile; sutures tear through native vessels. Operating on an undiagnosed vascular EDS patient is a recognized cause of intraoperative catastrophe.	~1 in 50,000–200,000. Most serious EDS subtype: median survival ~48 years. First major complication by age 40 in ~80%. Arterial rupture is most common cause of death. US: ~730–2,920 US total; ~100–400 US pediatric Global: ~29,000–116,000 global Source: Vascular EDS ~1/50,000–200,000; COL3A1 mutations; median survival ~48 years	Genetic testing (COL3A1 sequencing): gold standard. CT angiography of entire arterial tree: baseline mapping. MRI/MRA preferred over invasive angiography -- arterial puncture can cause uncontrollable hemorrhage. Echo: aortic root and valve. Avoid invasive arterial procedures unless absolutely necessary.	SOC: Conservative management strongly preferred — avoid elective surgery. Celiprolol (beta blocker with vasodilatory properties) — only agent with RCT evidence for reducing arterial event rates (BBEST trial). Surgical intervention for life-threatening rupture using modified technique (pledgeted sutures, Teflon felt, minimal tissue handling); endovascular preferred over open. Pre-operative COL3A1 screening before all vascular/cardiac surgery. Emerging: Losartan and irbesartan trials ongoing. Improved endovascular techniques for arterial complications (covered stents, fenestrated grafts). Genetic counseling: pregnancy carries very high risk (managed at specialized centers with MFM + cardiac).

VASCULAR AND METABOLIC CONDITIONS

#	CONDITION	DESCRIPTION	FREQUENCY	DIAGNOSIS	TREATMENT
55	Takayasu Arteritis	Large-vessel granulomatous vasculitis affecting the aorta and major branches (subclavian, carotid, renal, mesenteric, coronary arteries). Predominantly affects adolescent girls. Causes progressive	~1-2 per million per year overall. Onset in childhood (~10-15% of all Takayasu cases). Female predominance 8-9:1. Highest incidence in Asia, Latin America, and	Clinical: absent or diminished pulses, limb BP differential >10 mmHg, bruits, renovascular hypertension. Inflammatory markers (ESR, CRP, IL-6). CT angiography or MRA: vessel wall thickening,	SOC: High-dose corticosteroids for active disease. Steroid-sparing agents: methotrexate, azathioprine, mycophenolate mofetil. Surgical/endovascular intervention reserved for burnt-out disease in remission (PET-CT confirmed). Bypass grafting preferred over angioplasty for surgical revascularization. Emerging: Tocilizumab (IL-6 receptor

		arterial stenosis, occlusion, and aneurysm formation. Cardiac complications: aortic regurgitation, coronary ostial stenosis, renovascular hypertension, limb ischemia. Critically: in the active inflammatory phase, surgical intervention carries very high restenosis rates -- surgery must be deferred until disease is in remission.	South Asia. Coronary involvement in ~10-30% of pediatric cases. US: ~100–200 new US pediatric cases/yr; ~1,000–2,000 US total Global: ~100,000–200,000 new global cases/yr Source: Takayasu arteritis ~1–2/million/yr overall; ~10–15% onset in childhood	stenosis, occlusion, aneurysm -- gold standard. PET-CT: detects active vessel wall inflammation -- key for determining readiness for surgery. Echo: AR severity, coronary ostia.	antagonist) — best evidence for refractory/steroid-dependent pediatric Takayasu (used off-label in pediatrics; FDA-approved for adult GCA). TNF inhibitors (infliximab, adalimumab). Abatacept under investigation. PET-CT standardization for disease activity monitoring.
56	Pulmonary Arterial Hypertension (PAH) 	Intrinsic pulmonary vascular disease causing progressive arterial obliteration, elevated PA pressures, and right heart failure. Idiopathic or associated with CHD.	Idiopathic PAH: 15-50 per million. PAH-CHD much more common in pediatric cardiac populations. US: ~1,095–3,650 US pediatric cases; ~5,000–10,000 US total PAH all ages Global: ~42,000–140,000 global PAH (all ages) Source: Simonneau et al., JACC 2019; idiopathic PAH 15–50/million; pediatric PAH 0.7–4.4/million children/yr	Right heart catheterization (gold standard: mPAP $\geq$ 25 mmHg, PVR $\geq$ 3 Wood units). Echo. V/Q scan. Genetic testing (BMPR2). 6-minute walk test.	SOC: Targeted combination therapy: ERAs (bosentan, ambrisentan, macitentan) + PDE-5 inhibitors (sildenafil, tadalafil) $\pm$ prostanoids (epoprostenol IV, treprostinil SC/inhaled/IV, iloprost inhaled). Calcium channel blockers for vasoreactive patients (acute vasoreactivity testing-guided). Lung or heart-lung transplantation for refractory disease. Emerging: Sotatercept (Winrevair, Merck, FDA 2024) — activin signaling inhibitor (anti-ActRIIA); landmark STELLAR trial showed significant reduction in PVR and 6MWD; rapidly becoming add-on SOC. Ralinepag (oral IP receptor agonist) in Phase 3. AAV-BMPR2 gene therapy in preclinical development.
57	Eisenmenger Syndrome	End-stage consequence of large unrepaired left-to-right shunt in which chronically elevated pulmonary blood flow produces irreversible pulmonary vascular obstructive disease. Pulmonary vascular	Prevalence in high-income countries: ~1-9 per million (markedly decreased since widespread early repair). Large VSD/PDA: Eisenmenger	Clinical: central cyanosis, clubbing, polycythemia, reduced exercise tolerance, hemoptysis. Echo: bidirectional or right-to-left shunt flow; severe RV hypertension; RV hypertrophy and dilation. Right heart	SOC: Cardiac repair absolutely contraindicated once Eisenmenger physiology established. PAH therapy (palliative): bosentan (best Eisenmenger RCT evidence), macitentan, sildenafil, tadalafil; prostanoids for advanced disease. Phlebotomy only for severe hyperviscosity (Hct $>$ 65%). Avoid: pregnancy (maternal mortality 30–50%), dehydration, high

		resistance rises until it exceeds systemic, causing right-to-left shunting and systemic cyanosis. Once established, cardiac repair is permanently and absolutely contraindicated -- closing the defect removes the 'pop-off valve' and causes acute right heart failure and death. A preventable disease: develops only when a surgically correctable defect is not repaired in time.	physiology by age 1-2 years if unrepaired. Large ASD: typically develops in 3rd-4th decade. 10-year survival with modern PAH therapy: ~80% vs. ~40-50% historically. US: ~73–660 US total (markedly declined post-CHD repair era) Global: ~2,900–26,000 global (markedly declined in high-income countries) Source: Beghetti & Galie, Eur Heart J 2009; prevalence ~1–9/million in high-income countries now	catheterization (essential): quantify PVR; vasodilator testing (negative in Eisenmenger). CT pulmonary angiography. CBC: polycythemia (Hct often 55-75%).	altitude, iron deficiency. Heart-lung transplantation for selected deteriorating patients. Emerging: Sotatercept (Winrevair): SuMILE trial in Eisenmenger ongoing — anticipated to shift management paradigm. Macitentan + tadalafil combination trials. Riociguat (sGC stimulator) has been used off-label in Eisenmenger with some evidence.
58	Homozygous Familial Hypercholesterolemia (HoFH) 🧬🧬🧬	Mutations in both LDL receptor gene copies cause inability to clear LDL. Aggressive atherosclerosis from childhood; MI before age 20 without treatment.	1 in 300,000–1,000,000. Hundreds to a few thousand US patients. US: ~330–1,100 US total HoFH; ~2,000–5,000 including milder FH Global: ~13,000–43,000 global HoFH Source: Goldberg et al., JACC 2022; HoFH ~1/300,000–1,000,000; LDL >400–500 mg/dL from birth	Lipid panel (LDL 400–1,000+ mg/dL). Genetic testing (LDLR, APOB, PCSK9). Carotid/coronary imaging. Vascular ultrasound.	SOC: High-intensity statins + ezetimibe + PCSK9 inhibitors (evolocumab, alirocumab). LDL apheresis every 1–2 weeks. Lomitapide (MTP inhibitor) for HoFH. Emerging: Evinacumab (Evkeeza, Regeneron, FDA 2021) — ANGPTL3 inhibitor; approved for HoFH ≥12 years; significant LDL reduction independent of LDL receptor function. Inclisiran (siRNA, Leqvio) — trials ongoing for HoFH. LDL receptor gene therapy (Phase 1/2 trials).

#	CONDITION	DESCRIPTION	FREQUENCY	DIAGNOSIS	TREATMENT
59	Long QT Syndrome (LQTS) 	Inherited ion channel disorder causing prolonged cardiac repolarization and risk of torsades de pointes, syncope, and sudden cardiac death. Over 17 genetic subtypes; LQTS1 and LQTS2 account for >90% of genotyped cases.	~1 in 2,000 individuals. Responsible for ~3,500 sudden deaths annually in the US. US: ~36,500–73,000 US pediatric cases; ~165,000 US total Global: ~1,450,000–2,900,000 global Source: Schwartz et al., Circulation 2012; LQTS ~1/2,000 individuals; ~3,500 US sudden deaths/yr attributed to LQTS	12-lead ECG (QTc >=480ms diagnostic; 460-480ms borderline). Genetic panel. Exercise stress test. Schwartz score. Family screening.	SOC: Beta blockers (nadolol or propranolol preferred over metoprolol) — first-line for all LQTS types. ICD for high-risk patients (cardiac arrest survivors, recurrent syncope on therapy). Avoidance of QT-prolonging medications. Activity restriction by genotype (LQT1 most vulnerable to exercise, LQT2 to auditory triggers). Emerging: Mexiletine add-on for LQTS3 (late sodium current blocker — reduces QTc and events; now established practice). Left cardiac sympathetic denervation (LCSD) for refractory patients or those unable to tolerate ICD. Gene therapy and pharmacogenomic therapy for specific KCNQ1/KCNH2 variants in preclinical development.
60	CPVT (Catecholaminergic Polymorphic VT) 	Rare inherited channelopathy (RYR2 or CASQ2 mutations) causing bidirectional or polymorphic VT during adrenergic stimulation. Structurally normal heart. SCD is often the first manifestation.	~1 in 10,000. Accounts for ~15% of autopsy-negative sudden cardiac death in the young. US: ~7,300–14,600 US total; ~1,500–3,000 US pediatric Global: ~290,000–580,000 global Source: Priori et al., Circulation 2002; CPVT ~1/10,000; RYR2 mutations in ~60–70% of cases	Exercise stress test or epinephrine provocation -- characteristic bidirectional or polymorphic VT. Normal resting ECG and echo. Genetic testing (RYR2, CASQ2). Family screening.	SOC: Nadolol (preferred beta blocker — highest efficacy in CPVT). Flecainide add-on (reduces VT burden significantly — established adjunctive therapy). ICD for survivors of cardiac arrest or breakthrough VT/VF on therapy. Strict exercise restriction. Emerging: Left cardiac sympathetic denervation (LCSD) — reduces ICD shocks and VT episodes as adjunctive therapy. RyR2-targeted stabilizers (dantrolene, flecainide). Gene silencing of RYR2 gain-of-function variants using ASO/siRNA — preclinical.
61	Brugada Syndrome 	Sodium channelopathy (SCN5A mutations in ~20-30%) causing distinctive ECG pattern and risk of ventricular fibrillation, predominantly during rest or sleep. Structurally normal heart.	~1 in 2,000 globally; higher prevalence in Southeast Asia. Male predominance 8-10:1. US: ~36,500–73,000 US total; ~4,000–8,000 US	Type 1 Brugada ECG pattern (coved ST elevation >=2mm in V1-V2) -- spontaneous or sodium channel blocker-provoked. Genetic testing (SCN5A). Fever management critical.	SOC: ICD — only proven therapy for symptomatic patients (VF survivors, spontaneous sustained VT). Quinidine for electrical storm, VF storms, or ICD-ineligible patients. Aggressive fever management (antipyretics). Avoidance of sodium channel-blocking drugs. Emerging: Subcutaneous ICD (S-ICD)

			pediatric Global: ~1,450,000–2,900,000 global Source: Antzelevitch et al., Circulation 2005; Brugada ~1/2,000 globally; higher prevalence in SE Asia		preferred in younger patients without pacing indication. Epicardial ablation of the RVOT arrhythmogenic substrate (Brugada hotspot — Haïssaguerre protocol) — potentially curative in specialized centers. Sodium channel gain-of-function research guiding novel pharmacology.
62	Short QT Syndrome (SQTS) 🧬🧬🧬	Rare inherited channelopathy characterized by pathologically shortened cardiac repolarization (QTc <320-340ms) and risk of ventricular fibrillation and SCD. Caused by gain-of-function mutations in potassium channel genes (KCNQ1, KCNH2, KCNJ2 -- the opposite mutations from LQTS). SCA or VF may be the first manifestation, including in infants. Critically distinct from LQTS: opposite pathophysiology; beta blockers used in LQTS are ineffective or potentially harmful in SQTS.	Estimated prevalence <1 in 10,000; significantly underdiagnosed. All age groups including infants (recognized cause of sudden infant death). Approximately 70 families described in literature as of 2023. US: ~1,460–7,300 US total (significantly underdiagnosed) Global: ~58,000–290,000 global Source: Gollob et al., JACC 2011; SQTS <1/10,000; gain-of-function mutations in KCNH2, KCNQ1, KCNJ2	12-lead ECG: QTc <320ms (definitive); 320-340ms (borderline). Note: QT may appear near-normal at fast heart rates because QT does not shorten appropriately with rate increase (abnormal QT-RR relationship). Genetic panel (KCNQ1, KCNH2, KCNJ2, CACNA1C, CACNB2). EP study: short atrial/ventricular refractory periods, VF inducibility. Family screening mandatory.	SOC: ICD: mainstay for symptomatic patients — careful programming mandatory due to tall, peaked T-waves causing T-wave oversensing. Quinidine or hydroquinidine for ICD-ineligible patients (prolongs QT, reduces VF inducibility). Family screening with 12-lead ECG. Activity restriction pending ICD placement. Emerging: Disopyramide for SQTS2 (KCNQ1-GOF). Genetic phenotype-specific ion channel pharmacology under investigation. S-ICD with careful QRS/T-wave sensing optimization.

GENETIC SYNDROMES WITH CARDIAC INVOLVEMENT

#	CONDITION	DESCRIPTION	FREQUENCY	DIAGNOSIS	TREATMENT
63	Noonan Syndrome (Cardiac) 🧬🧬🧬	RASopathy (PTPN11 in ~50%). Cardinal cardiac features: pulmonary stenosis (50-60%), HCM (20-30%), ASD. HCM in Noonan is often refractory and may progress rapidly	~1 in 1,000-2,500 live births. Most common syndromic cause of CHD after Down syndrome. US: ~29,000–73,000 US total;	Clinical: short stature, typical facies, cryptorchidism. Genetic panel (RASopathy genes). Echocardiography. ECG (left axis deviation). Surveillance echo for	SOC: PS: balloon valvuloplasty (dysplastic valves have lower success rates than typical PS — may require surgical valvotomy). HCM: beta blockers, CCBs. Surgical myectomy for refractory LVOTO. ICD for high-risk HCM. Emerging: Trametinib (MEK inhibitor,

		in infancy.	~10,000–25,000 US pediatric Global: ~1,160,000–2,900,000 global Source: Romano et al., Eur J Med Genet 2010; Noonan ~1/1,000–2,500 births; PTPN11 ~50%	HCM progression.	Novartis) — FDA Breakthrough Designation for HRAS/RAS pathway-mutant Noonan HCM; active trials showing HCM regression. Broader RASopathy-targeted therapies (SHP2 inhibitors) for other Noonan variants. Mavacamten for Noonan HCM with LVOTO (off-label, emerging evidence).
64	Williams Syndrome / SVAS 	Contiguous gene deletion syndrome (7q11.23; elastin gene ELN). SVAS present in ~75% and is the leading cause of morbidity. Sudden death risk during cardiac procedures due to bilateral coronary ostial involvement.	~1 in 7,500 live births. US: ~29,000–43,000 new US cases/yr; ~870,000–1,300,000 US total Global: ~1,120,000–1,680,000 new global cases/yr Source: Morris et al., Pediatrics 2010; Williams syndrome ~1/7,500 live births; ELN deletion 7q11.23	Clinical diagnosis + FISH or chromosomal microarray for 7q11.23 deletion. Echo (SVAS gradient, coronary origins, branch PA stenosis). CT angiography for complex anatomy.	SOC: Surgical SVAS repair (patch aortoplasty — Doty or McGoon techniques) for significant gradients. Peripheral PA stenosis: transcatheter balloon dilation ± stenting. Mandatory pre-anesthesia cardiac evaluation due to SCD risk from coronary ostial involvement or bilateral PA stenosis. Lifelong cardiac surveillance. Emerging: Elastin augmentation strategies targeting ELN haploinsufficiency. Transcatheter SVAS approaches for discrete membranous lesions. Improved coronary ostial protection techniques perioperatively.
55b	22q11.2 Deletion Syndrome (Cardiac) 	Most common microdeletion syndrome (TBX1 haploinsufficiency). Cardiac defects in 75–80%; conotruncal anomalies predominate. Associated with immune deficiency, hypocalcemia, palatal abnormalities, and psychiatric illness.	~1 in 4,000 live births. Present in 50–70% of Interrupted Aortic Arch, ~15% of Truncus Arteriosus, ~15% of TOF.	FISH or SNP microarray (22q11.2 deletion). Echocardiography for conotruncal anatomy. Calcium levels. T-cell counts (DiGeorge immune deficiency).	SOC: Cardiac repair per specific defect (conotruncal anomaly — standard surgical management). Calcium supplementation for hypocalcemia. Thymic transplantation or T-cell reconstitution for complete DiGeorge syndrome with severe immune deficiency. Irradiated blood products if immunocompromised. Lifelong psychiatric and neurodevelopmental surveillance. Emerging: Partial thymus allograft transplantation (GT-DG, Enzyvant, FDA 2021) for complete DiGeorge syndrome — improved T-cell reconstitution. TBX1/RAS pathway targeted therapies in research. Prenatal diagnosis with fetal echo and genetic counseling for known carriers.
65	Alagille Syndrome (Cardiac) 	Autosomal dominant multisystem disorder	~1 in 30,000 live births.	Genetic testing (JAG1 or NOTCH2 sequencing).	SOC: Transcatheter balloon pulmonary arterioplasty and stenting: mainstay of

		(JAG1 mutations ~94%, NOTCH2 ~6%). Causes bile duct paucity, distinctive facial features, vertebral anomalies, posterior embryotoxon, and cardiovascular disease. Cardiac manifestation: diffuse peripheral pulmonary arterial stenosis (bilaterally, involving segmental and subsegmental branches) in ~90% of patients. PA stenosis is the leading cause of cardiac-related mortality. Liver transplantation does NOT resolve the cardiac manifestations.	Cardiovascular disease affects ~94%; clinically significant PA stenosis in ~90%. Cardiac complications (RV failure from severe PA stenosis) account for ~15% of deaths. US: ~120–180 new US cases/yr; ~4,000–6,000 US total. Global: ~4,700–7,000 new global cases/yr. Source: Alagille et al., J Pediatr 1987; Alagille ~1/30,000 births; JAG1/NOTCH2 mutations; cardiac involvement in ~94%	Clinical: cholestasis, butterfly vertebrae, posterior embryotoxon, triangular facies. Echo: RV pressure estimation. Cardiac MRI: quantitative bilateral PA blood flow, RV size and function. Cardiac catheterization + bilateral pulmonary angiography: gold standard for PA anatomy and pressures. Liver biopsy: bile duct paucity.	catheterization management; serial procedures throughout childhood targeting sub-systemic RV pressure. Cutting balloon and stent approaches for fibrotic lesions. Liver transplantation for biliary cirrhosis (~20–30%). Annual cardiac MRI and Doppler echo for RV pressure tracking. Emerging:Odevixibat (Bylvay, Albireo, FDA 2021) for cholestasis in Alagille syndrome — reduces pruritus and may slow liver disease progression. JAG1/Notch pathway modulation to promote PA growth (preclinical). Drug-eluting balloons for resistant PA stenosis.
66	CHARGE Syndrome 	Autosomal dominant multisystem developmental disorder caused by CHD7 mutations (~90% of cases). Acronym: Coloboma, Heart defects, Atresia choanae, Retardation of growth, Genital abnormalities, Ear anomalies with hearing loss. Cardiovascular malformations in ~75-80%; predominantly conotruncal: TOF (~20%), IAA, AVSD, VSD, PA-VSD, and complex heterotaxy. One of the most common causes of	~1 in 8,500-12,000 live births. Cardiac defects in ~75-80%; clinically significant CHD requiring intervention in ~60%. Conotruncal defects predominate (~40% of cardiac defects). US: ~300–420 new US cases/yr; ~10,000–15,000 US total. Global: ~11,700–16,500 new global cases/yr. Source: Zentner et	Clinical diagnosis (Blake or Verloes criteria). CHD7 gene sequencing: positive in ~90%. Echo: complete cardiac anatomy. CT angiography for complex arch anatomy. Brain MRI: hypoplastic/absent semicircular canals (pathognomonic for CHARGE -- present in >90%). Choanal atresia: CT of nasal passages. Ophthalmology, audiology, endocrine evaluation.	SOC:Cardiac repair: standard surgical management per specific defect. Sequential choanal atresia + cardiac surgery prioritization — choanal atresia repair first if causing neonatal respiratory distress. Anesthetic precautions: difficult airway, absent balance system, sensorineural hearing loss. Cochlear implantation for profound deafness. GH replacement for GH deficiency. Multidisciplinary management. Emerging:CHD7-pathway targeted therapies in preclinical development. Improved sensorineural hearing rehabilitation. Vestibular rehabilitation for absent semicircular canals.

		combined deaf-blindness in children.	al., Clin Genet 2010; CHARGE ~1/8,500–12,000 births; CHD7 de novo mutation in ~60–70%		
ARRHYTHMIAS					
#	CONDITION	DESCRIPTION	FREQUENCY	DIAGNOSIS	TREATMENT
67	Congenital Heart Block	Disruption of AV conduction present from birth. Complete form occurs in ~1-2% of pregnancies with maternal anti-Ro/SSA antibodies. Fetal complete heart block carries ~20-30% mortality without intervention.	~1 in 20,000 live births overall; 1-2% of pregnancies with anti-Ro/SSA antibodies. US: ~180–360 new US cases/yr; ~5,000–10,000 US total Global: ~7,000–14,000 new global cases/yr Source: Congenital heart block ~1/20,000 births; anti-Ro/SSA antibody in neonatal autoimmune form	Fetal heart rate monitoring. Fetal echocardiography. Postnatal: ECG. Maternal autoantibody testing.	SOC: Fetal: maternal dexamethasone or IVIG (limited efficacy for anti-Ro/SSA antibody-mediated block). Postnatal: permanent pacemaker implantation for symptomatic complete heart block or resting rate <55 bpm. Emerging: Hydroxychloroquine for anti-Ro/SSA antibody-positive mothers in subsequent pregnancies to prevent recurrence (PATCH trial data). In utero pacing — experimental. Beta-sympathomimetics (isoproterenol) as temporizing bridge in neonatal period.
68	Junctional Ectopic Tachycardia (JET)	Abnormal rhythm from the AV node. Postoperative JET occurs in ~1-5 per 1,000 pediatric cardiac surgeries. Congenital JET causes tachycardia-induced cardiomyopathy if rates remain persistently elevated.	Postoperative JET: 1-5 per 1,000 pediatric cardiac surgeries. Congenital JET: rare. US: ~1–10 new congenital JET US cases/yr; post-op JET ~500–2,500 US cases/yr Global: ~40–400 new congenital global cases/yr Source: Congenital JET very rare; postoperative JET ~1–5/1,000 pediatric	ECG (narrow complex tachycardia with AV dissociation). Holter for congenital JET.	SOC: Postoperative JET: amiodarone IV, induced hypothermia (cooling to 34–35°C), pacing above JET rate (overdrive atrial pacing). Dexmedetomidine sedation. Congenital JET: catheter ablation (80–90% success using cryoablation near His to reduce heart block risk). Emerging: Ivabradine (If channel blocker) for postoperative JET — emerging case series showing effective heart rate reduction without negative inotropy; particularly useful in low-output states. Cryoablation preferred over RF near His bundle for congenital JET.

			cardiac surgeries		
69	Ventricular Tachycardia (VT)	Rapid ventricular rhythm (≥ 3 beats, >120 bpm). May be idiopathic (benign RVOT-VT) or associated with structural disease, cardiomyopathy, or post-surgical scar.	Scar-related VT in repaired TOF: $\sim 0.2\%$ /year sudden death risk. SCA is presenting event in $\sim 25\text{--}30\%$. US: $\sim 2,000\text{--}5,000$ new US pediatric VT cases/yr (clinically significant) Global: $\sim 77,000\text{--}194,000$ new global cases/yr Source: Khairy et al., Circulation 2008; scar-related VT most prevalent in repaired TOF; $\sim 0.2\%$ /yr SCA risk	ECG. Holter/event monitor. Echocardiography. Cardiac MRI (scar assessment). EP study for catheter ablation planning.	SOC: ICD for high-risk structural VT or VF survivors. Catheter ablation for idiopathic VT (80–90% success — RVOT, fascicular, papillary muscle). Antiarrhythmics (amiodarone, mexiletine for scar-related VT). Treat underlying structural cause. Emerging: Stereotactic body radiotherapy (SBRT/cardiac radiosurgery) for refractory scar-related VT not amenable to catheter ablation — emerging as salvage therapy. Pulsed-field ablation (PFA) under investigation for VT substrate modification.
70	Supraventricular Tachycardia (SVT)	Most common significant pediatric arrhythmia. Rapid rhythm above ventricles — most commonly AVRT (accessory pathway) or AVNRT. WPW syndrome carries risk of pre-excited AF and sudden death.	1 in 250–1,000 children. WPW prevalence: 1–3 per 1,000 children. US: $\sim 73,000\text{--}292,000$ US pediatric cases Global: $\sim 2,900,000\text{--}11,600,000$ global pediatric cases Source: AHA pediatric SVT guideline 2016; SVT $\sim 1/250\text{--}1,000$ children; WPW $\sim 1\text{--}3/1,000$	ECG during episode. Adenosine for diagnosis and acute termination. Holter/event monitor. Delta wave on resting ECG identifies WPW.	SOC: Acute: vagal maneuvers, adenosine IV. Long-term: catheter ablation (90–95% cure rate for $>15\text{--}20$ kg). Definitive treatment for high-risk WPW (shortest pre-excited RR <250 ms) regardless of symptoms. Beta blockers or flecainide for suppression if ablation deferred. Emerging: Zero-fluoroscopy electroanatomical mapping (CARTO, EnSite) standard in pediatric centers — reducing radiation exposure. Cryoablation for para-Hisian and midseptal pathways — lower heart block risk than RF.
71	Ectopic Atrial Tachycardia (EAT)	Sustained/incessant tachycardia from abnormal atrial focus. Incessant nature causes tachycardia-induced cardiomyopathy in 25–50% of untreated cases.	$\sim 10\text{--}15\%$ of pediatric SVT cases. LV dysfunction reverses substantially after rate control or	ECG (P-wave morphology differs from sinus; incessant pattern). Holter. Echo for LV function assessment.	SOC: Antiarrhythmic agents (60–70% rate control): beta blockers, flecainide, sotalol, amiodarone. Catheter ablation preferred when accessible (80–90% success with high-density mapping). Beta blockers for suppression. Emerging: Ivabradine for rate control in

			ablation. US: ~7,300–10,950 US pediatric cases (10–15% of pediatric SVT) Global: ~290,000–437,000 global pediatric cases Source: EAT ~10–15% of pediatric SVT; automatic ectopic focus; LV dysfunction reverses with rate control		medically refractory EAT — off-label but effective without negative inotropy. High-density electroanatomical mapping (Ripple mapping, omnipolar technology) improving ablation success for focal AT.
72	Atrial Flutter / Fibrillation	Flutter: organized rapid atrial rhythm (250-350 bpm). AFib: chaotic atrial activity. In children, often CHD-related; flutter common in Fontan patients (15-20% by adulthood).	Neonatal flutter: 1 in 10,000-20,000 births. Fontan intraatrial reentrant tachycardia in ~15-20% by adulthood. US: ~1,800–3,600 new neonatal atrial flutter cases/yr; ~50,000–100,000 Fontan-IART US total Global: ~70,000–140,000 new global neonatal atrial flutter cases/yr Source: Neonatal atrial flutter ~1/10,000–20,000 births; intraatrial reentrant tachycardia (IART) in ~30–50% of Fontan patients by 10 yrs	ECG. Holter. Transesophageal echo for thrombus before cardioversion. EP study for ablation.	SOC:Acute: electrical cardioversion or chemical cardioversion (ibutilide, amiodarone). Catheter ablation: typical flutter >90% cure; CHD-related intra-atrial reentrant tachycardia (IART) 50–70%. Rate control: beta blockers, digoxin. Anticoagulation. Antiarrhythmics (sotalol, amiodarone). Emerging:Pulsed-field ablation (PFA — Farawave, Medtronic, FDA 2024) — non-thermal ablation for AFib; entering CHD population trials; may reduce risk of collateral injury to esophagus/phrenic nerve. High-density substrate mapping for complex CHD-IART.
73	Sick Sinus Syndrome (SSS)	Sinus node dysfunction causing pathological bradycardia, chronotropic incompetence, sinus	Post-operative SSS after Mustard/Senning: ~20-40% by 20	ECG: sinus bradycardia below age-appropriate lower limit, sinus pauses >3 seconds, prolonged	SOC:Permanent pacemaker (DDD/DDDR rate-responsive) for symptomatic SSS. Epicardial leads for small children or complex Fontan/Glenn anatomy.

		pauses, and tachy-brady syndrome. Two forms: (1) Congenital SSS -- associated with heterotaxy, L-TGA; (2) Post-operative SSS (more common) -- most common after Mustard/Senning, Fontan, ASD repair, and Maze procedures. Most common indication for permanent pacemaker implantation in the pediatric cardiac population. Tachy-brady syndrome is particularly dangerous: bradycardia causes syncope while tachyarrhythmia causes hemodynamic compromise and thromboembolism.	years post-op. Post-Fontan SSS: ~10-20%. Congenital SSS (heterotaxy): ~10-30% of heterotaxy patients. Overall most common indication for permanent pacemaker in pediatric CHD patients. US: ~3,000–8,000 post-Mustard/Senning SSS cases; ~5,000–15,000 post-Fontan SSS US total Global: ~115,000–310,000 global post-operative SSS Source: Post-op SSS after Mustard/Senning in ~20–40% by 20 yrs; post-Fontan SSS ~30–40%	post-tachycardia pause. Holter monitoring (24-72 hours or longer-term ILR): documents bradycardia-tachycardia alternation. Exercise stress test: chronotropic incompetence (failure to achieve >80% of maximum predicted heart rate). EP study: sinus node recovery time (SNRT).	Antiarrhythmics (sotalol, amiodarone) or catheter ablation for tachyarrhythmia component of tachy-brady. Anticoagulation for tachy-brady with structural heart disease. Emerging:Leadless pacemakers (Micra AV, Medtronic) for eligible adolescent/young adult patients. Conduction system pacing (His-bundle pacing, left bundle branch area pacing/LBBP) to preserve ventricular synchrony and reduce pacing-induced cardiomyopathy — expanding in pediatric practice.
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CARDIAC TUMORS

#	CONDITION	DESCRIPTION	FREQUENCY	DIAGNOSIS	TREATMENT
74	Cardiac Rhabdomyoma 	Most common primary cardiac tumor in children. Benign hamartoma of cardiac muscle; strongly associated with tuberous sclerosis complex (TSC) in 50-80%. Most regress spontaneously; large or multiple tumors cause obstruction, arrhythmia, or heart failure.	~1 in 100,000. Present in ~50% of TSC patients. Frequently detected on fetal echocardiography. US: ~360–720 new TSC-related rhabdomyoma cases/yr; ~730–1,460 US total Global: ~14,000–28,000 new global cases/yr	Fetal/neonatal echo (multiple echogenic masses). MRI for complex tumors. Genetic testing for TSC1/TSC2 mutations. EEG/MRI brain for TSC neurological involvement.	SOC:Observation — majority regress spontaneously in infancy. Everolimus (mTOR inhibitor, FDA-approved for TSC) for TSC-associated rhabdomyomas causing hemodynamic compromise or arrhythmia. Surgical resection only for life-threatening obstruction not responding to medical therapy. Emerging:Sirolimus as alternative mTOR inhibitor. Prenatal diagnosis with in utero everolimus treatment under compassionate use protocols for hydrops/critical obstruction. Genetic testing for TSC1/TSC2 to guide therapy and family counseling.

			Source: Cardiac rhabdomyoma ~1/100,000; ~50% of TSC patients; majority regress spontaneously		
75	Cardiac Fibroma 🧬	Second most common primary cardiac tumor in children. Benign fibrous tumor, non-regressing. Associated with Gorlin syndrome (PTCH1) in ~3%. Causes ventricular obstruction, arrhythmia, and sudden cardiac death.	Rare; <1 per 100,000. Significant SCD risk from VT/VF with large interventricular septal involvement. US: ~36–72 new US cases/yr; ~500–1,000 US total Global: ~1,400–2,800 new global cases/yr Source: Cardiac fibroma <1/100,000; significant SCD risk from VT/VF in large interventricular lesions	Echocardiography (large, echogenic, non-contractile intramyocardial mass). Cardiac MRI (central fibrous 'dark' appearance on LGE -- pathognomonic). Genetic testing for Gorlin syndrome (PTCH1).	SOC: Surgical resection (partial or complete) for large symptomatic tumors. ICD for VT/SCD prevention. Catheter ablation for arrhythmia arising from fibroma. Cardiac transplantation for unresectable tumors causing severe HF. Emerging: Hedgehog pathway inhibition (vismodegib for PTCH1-associated fibromas in Gorlin syndrome) under investigation. MRI-guided cardiac mapping to define arrhythmia substrate within fibroma. Cryoablation as adjunct to partial resection.
76	Cardiac Myxoma (Pediatric) 🧬	Most common benign cardiac tumor overall but rare in children. Gelatinous polypoid mass usually arising from the interatrial septum. Causes obstruction, systemic embolism, and constitutional symptoms. Carney complex (PRKAR1A mutation) associated in ~10% of pediatric myxomas.	Rare in children; <5% of cardiac myxomas occur in those under 15. US: ~10–18 new US pediatric cases/yr (<5% of all cardiac myxomas) Global: ~400–700 new global pediatric cases/yr Source: Cardiac myxoma most common benign cardiac tumor in adults; <5% in children under 15	Echocardiography (pedunculated mobile mass attached to fossa ovalis). Cardiac MRI. Genetic testing for PRKAR1A (Carney complex). Ophthalmology exam if Carney complex suspected.	SOC: Surgical excision with wide margins (and base of stalk) — curative. Carney complex: echo surveillance every 12 months lifelong due to high recurrence risk (~20%). Anticoagulation for embolic events. Emerging: PRKAR1A genetic screening for Carney complex family members. Minimally invasive/thoracoscopic excision for recurrent myxoma in selected anatomy. Improved imaging (cardiac MRI) for tumor characterization and pre-operative planning.
77	Intrapericardial Teratoma	Germ cell tumor arising within the pericardial	Most common cardiac/pericardial	Fetal echo: large pericardial effusion with	SOC: Fetal: pericardiocentesis for hydrops threatening fetal demise (temporizing).

		space, attached to the root of the aorta or pulmonary artery. Not intracardiac -- compresses the heart from the outside. The most common pericardial/cardiac tumor of the fetus and neonate and the most common cause of large fetal pericardial effusion. The effusion causes cardiac tamponade, fetal hydrops, and -- if untreated -- fetal or neonatal death. Histologically benign in the vast majority; postnatal surgical excision is curative.	tumor presenting in utero or neonatal period. Estimated ~1 in 200,000-400,000 live births. ~50% present as fetal hydrops from pericardial effusion. Recurrence after complete excision is rare. US: ~180–360 new US intrapericardial teratoma cases/yr Global: ~7,000–14,000 new global cases/yr Source: Most common cardiac/pericardial tumor presenting in utero or neonatal period	echogenic heterogeneous mass at the base of the great vessels -- virtually pathognomonic. Fetal MRI: mass anatomy and relationship to great vessel roots. Postnatal echo and CT for surgical planning. AFP: elevated in fetal and neonatal serum (germ cell origin).	Delivery planning at center with immediate postnatal surgical capability. Postnatal: complete surgical excision — curative; CPB often not required. ECMO for pre-operative hemodynamic instability. Adjuvant chemotherapy for malignant/immature teratoma (BEP regimen). Emerging:EXIT (ex utero intrapartum treatment) procedure for large tumors causing airway compromise at delivery. Fetal MRI for precise tumor characterization and surgical planning. Minimally invasive resection for smaller, extrapericardially accessible tumors.
78	Cardiac Rhabdomyosarcoma	Most common primary malignant cardiac tumor in children. Arises from cardiac striated muscle (most commonly ventricular myocardium or AV valves) and is locally invasive. Unlike benign pediatric cardiac tumors, rhabdomyosarcoma is aggressive with local invasion of cardiac structures and hematogenous metastasis. There is no spontaneous regression. Complete surgical resection is rarely achievable due to local invasion.	Rare; primary cardiac rhabdomyosarcoma accounts for ~25% of primary malignant cardiac tumors. Pediatric cardiac malignancies overall: ~0.001-0.003% of childhood cancers. Prognosis is poor: 5-year survival <20%. US: ~10–25 new US primary cardiac rhabdomyosarcoma cases/yr Global: ~400–1,000 new global cases/yr Source: Primary cardiac	Echo: large, irregular, invasive-appearing cardiac mass -- often infiltrates rather than displaces normal structures (unlike benign tumors). Cardiac MRI: tissue characterization, extent of invasion, pericardial involvement. CT chest/abdomen/pelvis for staging. PET scan for systemic staging. Tissue biopsy: essential for diagnosis and histological subtype.	SOC:Multidisciplinary management (cardiac surgery + pediatric oncology). Surgical debulking to relieve obstruction and obtain tissue; complete resection rarely possible. Adjuvant chemotherapy: VAC regimen (vincristine, actinomycin D, cyclophosphamide) — standard rhabdomyosarcoma protocol. Radiation therapy for local control. Palliative intent frequently the realistic goal. Emerging:Pembrolizumab (anti-PD-1) + chemotherapy combinations in pediatric rhabdomyosarcoma trials. CDK4/6 inhibitors and MDM2 antagonists for specific genomic subsets. CAR-T cell approaches targeting RMS-associated antigens in early trials.

			rhabdomyosarcoma ~25% of primary malignant cardiac tumors; extremely rare		
INFLAMMATORY / ACQUIRED					
#	CONDITION	DESCRIPTION	FREQUENCY	DIAGNOSIS	TREATMENT
79	Kawasaki Disease	Acute self-limited vasculitis preferentially targeting the coronary arteries. Leading cause of acquired heart disease in children in developed countries. Coronary artery aneurysms develop in 20-25% of untreated children.	9-20 per 100,000 children under 5 in the US; significantly higher in Japan/East Asia (135 per 100,000). ~4,200-5,000 US cases annually. US: ~72,000–160,000 new US cases/yr (children under 5); ~4,500–9,000 coronary aneurysm cases/yr Global: ~1,400,000–3,100,000 new global cases/yr Source: McCrindle et al., AHA Circulation 2017; 9–20/100,000 US children under 5; Japan ~300/100,000 (highest globally)	Clinical diagnosis: fever ≥ 5 days plus ≥ 4 of 5 classic features (rash, conjunctival injection, mucous membrane changes, extremity changes, cervical lymphadenopathy). Echo at diagnosis, 2 weeks, and 6-8 weeks for coronary assessment.	SOC: High-dose IVIG (2 g/kg) + aspirin within first 10 days (reduces aneurysm risk to $<5\%$). Corticosteroids as adjunctive therapy or for IVIG-resistant cases. Long-term anticoagulation (warfarin + low-dose aspirin) for giant aneurysms. Surveillance catheterization for giant aneurysms with stenosis. Emerging: Infliximab (TNF inhibitor) for IVIG-resistant KD — comparable efficacy to second IVIG dose with fewer infusion reactions; now widely used at many centers. Cyclosporine for refractory KD. Anakinra (IL-1Ra) for severe refractory cases. Predictive scoring tools (Kobayashi, Egami) to identify IVIG resistance risk pre-treatment.
80	MIS-C (Multisystem Inflammatory Syndrome in Children)	Post-infectious hyperinflammatory syndrome occurring 2-6 weeks after SARS-CoV-2 infection. Causes myocarditis, coronary artery dilation, ventricular dysfunction, and shock.	Estimated 1 in 3,000-4,000 SARS-CoV-2-infected children. Over 9,000 cases reported in the US through 2023. US: ~3,500–5,000 new US cases during 2020–2023 peak; ~600–	CDC case definition: fever + elevated inflammatory markers + multiorgan involvement + SARS-CoV-2 evidence. Echo (LV dysfunction, coronary artery Z-scores). Troponin, BNP. Inflammatory markers (CRP, ferritin, D-dimer).	SOC: IVIG (2 g/kg) + aspirin. Corticosteroids for refractory or severe cases. Hemodynamic support: inotropes, ECMO for severe cardiogenic shock. Anticoagulation for coronary dilation (z-score ≥ 2.5) or significant LV dysfunction. Emerging: Anakinra (IL-1 receptor antagonist) for refractory MIS-C with macrophage activation syndrome features. Standardized cardiac MRI follow-up protocols to assess myocardial recovery.

			1,000/yr from 2023 onward Global: ~50,000–150,000 total global cases reported (CDC, WHO surveillance through 2024) Source: Feldstein et al., NEJM 2020; CDC MIS-C surveillance; incidence markedly decreased with Omicron variant + vaccination		Long-term data on coronary outcomes and myocardial fibrosis accumulating.
81	Infective Endocarditis (IE)	Infection of the endocardial surface of the heart, predominantly affecting cardiac valves. In children, predominantly affects those with underlying CHD (VSDs, AVSDs, TOF, prosthetic valves, indwelling lines). Organisms: Streptococcus viridans group (most common), Staphylococcus aureus (most aggressive).	~0.3-0.5 per 100,000 children/year overall; ~5-10 per 1,000 person-years in high-risk CHD. Mortality 5-10% with appropriate treatment. US: ~215–360 new US pediatric cases/yr; ~10,000–20,000 in high-risk CHD populations/yr Global: ~8,400–14,000 new global pediatric cases/yr Source: Baltimore et al., Circulation 2015; IE ~0.3–0.5/100,000 children/yr in structurally normal hearts; 5–10/1,000 person-yrs in high-risk CHD	Duke Criteria (modified 2000). Echocardiography (TTE first, TEE for prosthetic valves). Blood cultures x3 before antibiotics.	SOC:Prolonged IV antibiotics (4–6 weeks; organism-directed per blood culture sensitivities). Empirical anti-staphylococcal coverage until cultures available (vancomycin or daptomycin). Early surgical consultation for: HF from valvular destruction, persistent bacteremia, large vegetations (>10 mm), perivalvular abscess. Dental prophylaxis for high-risk CHD. Emerging:Bacteriophage therapy for prosthetic valve IE caused by resistant organisms (compassionate use, case series). Biofilm-targeted antibiotics (daptomycin + rifampin combinations). Updated AHA/ACC 2023 IE guidelines: dental prophylaxis indications refined; broader cardiac device IE management.
82	Rheumatic Heart Disease (RHD)	Autoimmune valvular damage triggered by repeated Group A	500-1,000 per 100,000 school-age children in endemic	Clinical history of rheumatic fever. Echocardiography per	SOC:Secondary prophylaxis: monthly benzathine penicillin G for 10 years or until age 25 (whichever longer) to prevent

		streptococcal pharyngitis. Leading cause of acquired heart disease in children globally. Mitral valve affected in >80% of cases.	regions. ~300,000 deaths annually worldwide. US: ~5,000–25,000 US pediatric cases/yr (predominantly immigrant/refugee populations) Global: ~500,000–1,000,000 new global cases/yr; ~300,000 deaths/yr Source: WHO GHO; RHD ~500–1,000/100,000 school-age in endemic regions; essentially rare in high-income countries	World Heart Federation criteria. Throat culture or rapid strep test for acute GAS pharyngitis.	recurrent GAS pharyngitis and rheumatic fever. Surgical valve repair or replacement for advanced valvular disease. Emerging: Group A streptococcal (M-protein based) vaccine — multiple candidates in Phase 2/3 trials; would be transformative for RHD prevention globally. Transcatheter mitral valve repair (MitraClip) for severe RHD-MR in resource-limited settings where surgery unavailable.
83	Acute Pericarditis	Inflammation of the pericardium causing chest pain, pericardial friction rub, and characteristic ECG changes (diffuse ST elevation, PR depression). Most commonly viral. Can progress to pericardial effusion and tamponade.	~1-2 per 100,000 children/year. Recurrent pericarditis occurs in 15-30% of cases. US: ~730–1,460 new US pediatric cases/yr Global: ~28,000–56,000 new global pediatric cases/yr Source: Imazio et al., Circulation 2007; ~1–2/100,000 children/yr; recurrent pericarditis in 15–30%	ECG (diffuse saddle-shaped ST elevation, PR depression). Echo (pericardial effusion, rule out tamponade). Cardiac MRI (pericardial inflammation on LGE). Troponin (elevated if myopericarditis). CRP, ESR.	SOC: NSAIDs (ibuprofen 600 mg TID or aspirin) — first-line for 2–4 weeks. Colchicine add-on (reduces recurrence by ~50%; COPE and ICAP trials). Corticosteroids for NSAID-refractory or autoimmune pericarditis. Exercise restriction for 3 months. Treat underlying cause. Emerging: Rilonacept (Arcalyst, Regeneron, FDA 2021) — IL-1 α/β inhibitor; approved for recurrent pericarditis refractory to colchicine (RHAPSODY trial). Anakinra (IL-1 receptor antagonist) for colchicine-refractory and IL-1-driven recurrent pericarditis; particularly in pediatrics. Pericardiectomy for constrictive pericarditis (see entry 84).
84	Constrictive Pericarditis	Progressive fibrosis and calcification of the pericardium that encases and constricts the heart, impairing diastolic ventricular filling. In	~5-10% of children with acute pericarditis develop some degree of constriction. Post-cardiac surgery	Echo: septal bounce (pathognomonic -- interventricular septal shift with respiration), biatrial enlargement, normal LVEF, annulus reversus	SOC: Surgical pericardiectomy (radical pericardial decortication of both ventricles, visceral and parietal) — definitive and curative. Complete pericardiectomy essential for lasting hemodynamic benefit. Anti-inflammatory therapy (NSAIDs +

		children, most commonly follows viral pericarditis, post-cardiac surgery (post-pericardiotomy syndrome), or TB pericarditis in endemic regions. Critically important distinction from restrictive cardiomyopathy (RCM, #38): constrictive pericarditis is surgically curable; RCM is not. The two conditions produce identical clinical presentations, requiring catheterization and MRI for differentiation.	constrictive pericarditis: ~0.2-0.3% of pediatric cardiac surgeries. TB pericarditis is the leading cause in sub-Saharan Africa. US: ~73–146 new US pediatric cases/yr Global: ~2,800–5,600 new global pediatric cases/yr Source: 5–10% of children with acute pericarditis develop some degree of constriction	(medial mitral annular e' > lateral e' on tissue Doppler). CT: pericardial thickening (>4mm) and calcification. Cardiac MRI: pericardial thickness, LGE, ventricular interdependence. Cardiac catheterization (gold standard): equalization of diastolic pressures; dip-and-plateau ('square root sign'); discordant ventricular pressure changes with respiration (key differentiator from RCM).	colchicine) for early/effusive-constrictive disease — ~20–30% resolve without surgery. Antituberculous therapy mandated for TB pericarditis. Diuretics for venous congestion while awaiting surgery. Emerging: Video-assisted thoracoscopic pericardiectomy (VATS) for selected cases — reduced morbidity vs. median sternotomy/thoracotomy. Cardiac MRI with LGE to identify pericardial fibrosis/inflammation and predict surgical benefit vs. spontaneous resolution.
85	Aortic Valve Regurgitation (Acquired/Post-Interventional)	Progressive regurgitation of the aortic valve resulting from congenital valve disease, prior balloon valvuloplasty, prior surgical valvotomy, infective endocarditis, rheumatic disease, or Ross procedure failure. A leading reason for re-intervention in patients with prior LVOT disease.	Affects ~20-40% of patients who underwent neonatal/infant aortic balloon valvuloplasty within 10-15 years. AR after Ross procedure conduit degeneration: ~15-20% by 10 years. US: ~5,000–12,000 US prevalent pediatric cases (post-intervention aortic regurgitation) Global: ~195,000–470,000 global Source: Post-balloon valvuloplasty AR in 20–40% of neonatal/infant aortic interventions	Echo: AR jet width, vena contracta, holodiastolic reversal in descending aorta. Cardiac MRI: gold standard for AR quantification (regurgitant volume, regurgitant fraction) and LV end-systolic volume index (LVESVi -- intervention threshold $\geq 25 \text{ mL/m}^2$). Serial surveillance echo every 6-12 months.	SOC: Surveillance to LV thresholds (LVESVi $\geq 25 \text{ mL/m}^2$, LVEF <50%, or symptoms). Ross or Ozaki procedure preferred in children and young adults. Mechanical valve replacement when repair criteria not met. TAVR valve-in-valve for failed bioprosthetic aortic valves in adolescents/adults. Emerging: Ozaki procedure (aortic valve neocuspidization with autologous pericardium) gaining broad adoption as Ross alternative — avoids pulmonary autograft reoperation. BASILICA technique (pre-TAVR leaflet laceration) to prevent coronary occlusion during valve-in-valve TAVR. Transcatheter aortic valve repair approaches in development.

#	CONDITION	DESCRIPTION	FREQUENCY	DIAGNOSIS	TREATMENT
86	Post-Cardiotomy Syndrome / Low Cardiac Output (LCOS)	Systemic inflammatory and hemodynamic response following cardiopulmonary bypass (CPB), manifesting as clinically significant low cardiac output (CI <2.0 L/min/m ²). Caused by myocardial ischemia-reperfusion injury, complement activation from bypass circuit contact, capillary leak, and post-operative systemic vasoconstriction. The leading perioperative cause of CICU morbidity and mortality in pediatric cardiac surgery.	25-35% of post-cardiotomy patients experience clinically significant LCOS in first 6-18 hours. Highest risk: neonatal operations (Norwood, arterial switch), prolonged bypass time, pre-operative ventricular dysfunction, deep hypothermic circulatory arrest. US: ~8,750–12,250 new US pediatric post-cardiotomy LCOS cases/yr Global: ~350,000–490,000 new global cases/yr Source: Hoffman et al., Semin Thorac Cardiovasc Surg 2003; LCOS in 25–35% of post-cardiotomy patients in first 24 hrs	NIRS (near-infrared spectroscopy) -- continuous regional O ₂ saturation; falling cerebral NIRS is an early LCOS indicator. Arterial line (continuous BP, blood gas access). Central venous line (filling pressures). Mixed venous SvO ₂ or SvcO ₂ (target >65-70%). Cardiac index measurement. Lactate trends (rising lactate = early LCOS indicator). Echo (LVEF, RV function, tamponade, residual lesions).	SOC: Optimize preload (CVP target 8–12 mmHg). Milrinone (first-line — prophylactic milrinone post-neonatal surgery reduces LCOS incidence by 64%, PRIMACORP trial). Epinephrine, dopamine for refractory cases. Vasopressin for vasodilatory shock. ECMO rescue for refractory LCOS (VA configuration; survival 40–60%). Neuromuscular blockade and sedation to reduce O ₂ demand. Emerging: Levosimendan (calcium sensitizer/K-ATP channel opener) as alternative/adjunct to milrinone — emerging evidence for lower risk of tachyarrhythmia vs. catecholamines. Terlipressin (V1a agonist) for vasodilatory shock post-CPB in pediatric centers.
87	Pulmonary Hypertensive Crisis (Post-Operative)	Sudden, severe elevation in pulmonary vascular resistance causing acute RV failure and hemodynamic collapse. A distinct life-threatening event from chronic PAH. Occurs in the early post-operative period in patients with CHD and pre-existing pulmonary hypertension. Triggered by hypoxia, hypercarbia, acidosis, suctioning, pain, or stimulation.	Occurs in ~5-10% of patients with pre-existing elevated PVR undergoing CHD surgery. Mortality from acute PH crisis: 15-30% per event. Most common in AVSD, large VSD, truncus arteriosus, TAPVR with delayed repair, and patients with Down syndrome. US: ~500–1,250	Clinical: sudden acute desaturation, hypotension, bradycardia, loss of pulse pressure. PA line: acute rise in PA pressures approaching or exceeding systemic systolic. NIRS: acute drop in cerebral saturation. Echo: acute RV dilation, septal shift, reduced PA flow. ABG: hypoxia, hypercarbia, acidosis.	SOC: Hyperventilation (pCO ₂ target 30–35 mmHg). 100% FiO ₂ . Inhaled nitric oxide (20–40 ppm) — first-line pharmacological therapy. IV sildenafil for iNO transition/weaning. Alkalinization (sodium bicarbonate). Deep sedation + neuromuscular blockade. Minimize stimulation. Prostacyclin (inhaled or IV) for iNO-refractory cases. ECMO for refractory PH crisis with hemodynamic collapse. Emerging: Inhaled treprostinil as alternative/adjunct to iNO for post-operative PH. Transition protocols from iNO to oral/inhaled PAH therapy (sildenafil, bosentan) to facilitate safe weaning.

			new US pediatric post-op PHC cases/yr Global: ~19,500–48,750 new global cases/yr Source: Occurs in ~5–10% of patients with pre-existing elevated PVR undergoing CHD surgery		Riociguat (sGC stimulator) for post-operative PH in adults.
88	Cardiac Tamponade	Pericardial fluid accumulation causing external cardiac compression, impaired ventricular filling, and obstructive cardiogenic shock. In the CICU, most commonly post-operative. Also occurs in myopericarditis, pericarditis, malignancy, and post-cardiac procedure.	Post-cardiac surgery tamponade: ~1-5% of pediatric cardiac surgeries. Late tamponade (post-pericardiotomy syndrome): ~10-30% of cardiac surgery patients. Cardiac catheterization-related tamponade: ~0.1-0.5% of procedures. US: ~500–2,500 new US pediatric cardiac tamponade cases/yr (post-cardiac surgery) Global: ~19,500–97,500 new global cases/yr Source: Post-cardiac surgery tamponade in ~1–5% of pediatric cardiac surgeries	Beck's triad: hypotension + elevated JVP + muffled heart sounds (late and insensitive). Echo: pericardial effusion + RA/RV collapse in diastole + plethoric IVC + respiratory variation in mitral/tricuspid inflow >25% (pathognomonic). Pulsus paradoxus: BP drop >10 mmHg on inspiration. CXR: enlarged cardiac silhouette.	SOC: Emergency ultrasound-guided pericardiocentesis (subxiphoid or apical approach). Surgical drainage for: post-operative hemorrhagic tamponade, posterior effusions not accessible by needle, clotted effusions. IV fluid bolus as temporizing bridge. Avoid positive pressure ventilation (reduces venous return, worsens tamponade). Emerging: Point-of-care ultrasound (POCUS) for real-time guidance becoming standard. Percutaneous balloon pericardiotomy for recurrent malignant effusions. Pericardioscopy-guided pericardiocentesis for complex or loculated effusions.
89	Chylothorax	Accumulation of chyle (lymphatic fluid rich in triglycerides and lymphocytes) in the pleural space due to injury to or	Occurs in 1-3% of pediatric cardiac surgeries overall; higher rates after Glenn (5-10%) and	Pleural fluid analysis: triglycerides >110 mg/dL (diagnostic). Lymphocyte predominance >80%. Daily chest tube output	SOC: Conservative: MCT diet, NPO with TPN for high-output chylothorax. Octreotide (somatostatin analog) IV or SQ. Chest tube drainage. Thoracic duct embolization (Itkin technique via cisterna chyli, 80–90%

		obstruction of the thoracic duct during cardiac surgery. Most common after Glenn and Fontan procedures, arch surgery, BT shunt, and mediastinal dissection.	Fontan procedures. Most commonly left-sided. US: ~500–1,500 new US pediatric chylothorax cases/yr Global: ~19,500–58,500 new global cases/yr Source: Chylothorax in 1–3% of pediatric cardiac surgeries overall; higher rates after Glenn/Fontan (5–8%)	tracking. Chest radiograph/ultrasound. MR lymphangiography for thoracic duct anatomy and leak localization.	closure) — increasingly preferred over surgery. Surgical thoracic duct ligation for embolization failures. Pleurodesis for recalcitrant cases. Emerging: Sirolimus (mTOR inhibitor) for chylous effusions — particularly Fontan-related chylothorax and lymphangi leiomyomatosis-related; active clinical use at many centers. Intranodal lymphangiography + embolization refining technique.
90	Diaphragm Paralysis (Phrenic Nerve Injury)	Injury to the phrenic nerve during cardiac surgery causing unilateral or bilateral diaphragmatic paralysis. Left phrenic nerve most commonly affected due to its proximity to the aortic arch, left PA, and ductus arteriosus dissection. Results in paradoxical diaphragmatic movement, reduced lung volumes, atelectasis, and prolonged ventilator dependence.	Occurs in 1-5% of pediatric cardiac surgeries. Higher rates in neonatal and infant operations and after arch surgeries. US: ~500–2,500 new US pediatric phrenic nerve injury cases/yr Global: ~19,500–97,500 new global cases/yr Source: Phrenic nerve injury in 1–5% of pediatric cardiac surgeries; higher rates in neonatal/infant operations	Difficulty weaning from mechanical ventilation without explanation. Ultrasound: bedside assessment of diaphragm motion -- gold standard; paradoxical motion is diagnostic. CXR: elevated hemidiaphragm. Fluoroscopy: sniff test.	SOC: Trial of non-invasive ventilation (BiPAP/CPAP) post-extubation (positioning affected side up, chest physiotherapy). Surgical diaphragm plication: definitive treatment for failed extubation; thoracoscopic approach preferred; after 4–6 weeks conservative trial (or immediately for bilateral paralysis). Emerging: Robot-assisted thoracoscopic diaphragm plication — reduced morbidity, comparable outcomes. Diaphragm pacing (phrenic nerve stimulation) for bilateral paralysis in selected cases. Ultrasound-guided assessment of diaphragm excursion to monitor recovery before committing to plication.
91	Protein-Losing Enteropathy (PLE) in Fontan	Life-limiting complication of Fontan circulation in which protein (predominantly albumin) leaks from the gut into the intestinal lumen due to chronically elevated systemic venous pressure	Affects 3-15% of Fontan patients; onset typically 2-10 years after Fontan completion. US: ~450–2,250 US Fontan-PLE total cases	Clinical: peripheral edema, ascites, pleural effusions, facial edema, diarrhea. Serum albumin <3.5 g/dL. Alpha-1-antitrypsin clearance. 99mTc-labeled albumin nuclear scan -- localizes	SOC: High-protein, fat-restricted (MCT fat) diet. Sodium and fluid restriction. Spironolactone + diuretics. Subcutaneous UFH (100 units/kg/day). Sildenafil. Budesonide (oral). Albumin infusions for acute decompensation. Catheter-created Fontan fenestration to decompress systemic venous pressure. Heart transplantation for

		and lymphatic hypertension. Results in hypoalbuminemia, generalized edema, ascites, pleural effusions, and immune deficiency. Five-year survival after PLE diagnosis is ~50%.	Global: ~17,500–87,500 global Fontan-PLE Source: Affects 3–15% of Fontan patients; onset typically 2–10 years after Fontan completion	gut protein loss. Cardiac catheterization for hemodynamic assessment. MR lymphangiography.	medically refractory PLE. Emerging: Thoracic lymphatic embolization (Fontan-specific lymphatic intervention — Itkin/CHOP protocol): emerging as major catheter-based intervention for PLE and plastic bronchitis; directly decompresses lymphatic hypertension.
92	Plastic Bronchitis in Fontan	Rare but potentially fatal complication of Fontan circulation in which bronchial casts composed of lymphatic fluid and inflammatory cells fill and obstruct the bronchial tree. Patients expectorate rigid, branching casts that replicate the bronchial anatomy. Pathophysiology: lymphatic hypertension from elevated Fontan venous pressure causes lymphatic fluid to leak into the airways.	Prevalence in Fontan population: ~1-4%. Mortality per acute episode: ~10-15%. Recurrent episodes are the rule rather than the exception. US: ~100–400 US total Fontan plastic bronchitis cases Global: ~3,900–15,600 global Source: Prevalence ~1–4% in Fontan population; mortality per acute episode ~10–15%	Clinical: acute respiratory distress, hypoxemia, expectorated or bronchoscopically retrieved branching casts (diagnostic). CT chest: branching hyperdense opacities filling the bronchi. Bronchoscopy: direct visualization and retrieval of bronchial casts. Echo/catheterization: Fontan hemodynamic assessment. MR lymphangiography.	SOC: Acute: bronchoscopy for cast removal. Inhaled tPA (tissue plasminogen activator) — standard of care at experienced centers; dissolves fibrin casts and reduces recurrence. Chest physiotherapy + airway clearance. Inhaled heparin (reduces cast formation). Optimize Fontan hemodynamics: sildenafil, bosentan. Heart transplantation for medically refractory plastic bronchitis. Emerging: Thoracic lymphatic embolization — same as PLE; increasingly effective for plastic bronchitis via lymphatic decompression. Inhaled dornase alfa as mucolytic adjunct. Combined heart-liver transplantation for cases with advanced FALD.
93	Fontan-Associated Liver Disease (FALD)	Progressive hepatic fibrosis, cirrhosis, and hepatocellular carcinoma resulting from chronic elevation of systemic venous pressure transmitted to the hepatic venous system in Fontan physiology. Nearly universal in long-term Fontan survivors. A leading cause of late morbidity and mortality in Fontan patients.	Hepatic fibrosis: virtually all Fontan patients within 10 years. Cirrhosis: ~50% within 10-15 years. Hepatocellular carcinoma (HCC): increasingly recognized; median age at HCC diagnosis ~30 years. US: ~4,500–9,000 US total Fontan-FALD (hepatic fibrosis)	Liver function tests (often normal until late disease; elevated GGT may be early marker). Serum elastography (FibroScan) -- non-invasive fibrosis staging. MRI liver: fibrosis, nodularity, HCC surveillance. AFP (HCC marker) -- annual. Liver biopsy: gold standard (used selectively). Hepatology co-management from 5-10 years post-Fontan.	SOC: No disease-modifying therapy proven to prevent or reverse FALD. Optimize Fontan hemodynamics to reduce hepatic venous pressure. Avoid hepatotoxic medications. HCC surveillance: 6-monthly AFP + annual liver MRI from 10 years post-Fontan. HCC treatment: resection or ablation if isolated. Combined heart-liver transplantation for advanced hepatic fibrosis + Fontan failure. Emerging: Non-invasive liver fibrosis staging (FibroScan/transient elastography, MR elastography) to monitor progression without biopsy. Antifibrotic agents (pirfenidone, nintedanib) under investigation for FALD. FIB-4 score and hepatic venous pressure gradient as surveillance markers.

			Global: ~175,000–350,000 global Fontan-FALD Source: Hepatic fibrosis present in virtually all Fontan patients within 10 years; cirrhosis ~50% by 20 years		
94	Fontan Failure	Progressive deterioration of Fontan physiology characterized by falling cardiac output, rising Fontan pressures, and multiorgan dysfunction. Clinical syndrome: low cardiac output, ascites, edema, pleural effusions, PLE, plastic bronchitis, FALD, and arrhythmias. The final common pathway for all Fontan patients if they survive long enough.	Estimated 40-50% of Fontan patients develop significant Fontan failure by 40 years of age. Annual Fontan failure rate approximately 1-2% per year after initial Fontan completion. US: ~7,200–14,400 US Fontan patients at risk (40–50% by age 40) Global: ~280,000–560,000 global Fontan patients at risk Source: Estimated 40–50% of Fontan patients develop significant failure by age 40	Echocardiography (ventricular function, AV valve regurgitation, Fontan pathway obstruction). Cardiac catheterization (Fontan pressure, PVR, cardiac output -- essential before transplantation listing). MRI (ventricular volumes and function). Liver assessment (FALD staging). PLE evaluation. Renal function. Functional assessment: 6-minute walk test, VO2 max.	SOC: Medical optimization: diuretics, afterload reduction, pulmonary vasodilators (sildenafil, bosentan). Rhythm management: antiarrhythmics, catheter ablation for IART. Catheter interventions: Fontan fenestration, Fontan pathway stenting for conduit obstruction, APC coil embolization. Fontan conversion surgery for failing atriopulmonary connections. Heart transplantation — the only definitive therapy. Combined heart-liver transplantation for advanced FALD. Emerging: Mechanical cavopulmonary assist (percutaneous TCPC pump — Fontan mechanical assist device trials). Thoracic lymphatic embolization for PLE/plastic bronchitis. Fenestration optimization with adjustable devices.
95	Acute Cellular Rejection (Post-Cardiac Transplant)	T-cell-mediated immune response against the transplanted heart. The primary cause of morbidity and death in the first year after cardiac transplantation. Graded by the ISHLT rejection grading system (0-3R based on endomyocardial biopsy histology).	Treated rejection episodes occur in 20-40% of pediatric recipients in the first year post-transplant. Severe rejection (Grade 2R or 3R): ~10-15% of patients in year 1. Adolescent non-adherence is the leading cause of	Endomyocardial biopsy (EMB): gold standard for diagnosis and grading (ISHLT grade 1R = mild; 2R = moderate; 3R = severe). EMB protocol: biweekly for first 2 months, monthly until year 1, then annually. Echo: LV dysfunction, new wall motion abnormality. Biomarkers: troponin,	SOC: Grade 1R: optimize oral immunosuppression; close follow-up. Grade 2R or 3R: IV methylprednisolone pulse (10–30 mg/kg/day × 3 days, max 1 g/day). Grade 3R refractory to steroids: rabbit ATG. AMR: plasmapheresis + IVIG + rituximab. Hemodynamic support: inotropes, ECMO as bridge to recovery. Maintenance immunosuppression optimization. Emerging: Bortezomib (proteasome inhibitor) for refractory AMR — reduces plasma cell burden. C1-esterase inhibitor

			late rejection. US: ~400–800 new US pediatric transplant rejection episodes/yr Global: ~15,000–30,000 global Source: Treated rejection episodes in 20–40% of pediatric recipients in first year post-transplant	BNP. Gene expression profiling (AlloMap).	(Berinert) for complement-mediated AMR. Belatacept (CTLA4-Ig) as calcineurin inhibitor alternative. Non-invasive rejection surveillance: donor-derived cell-free DNA (dd-cfDNA) as liquid biopsy.
96	Primary Graft Dysfunction (PGD)	Severe cardiac dysfunction of the transplanted heart occurring within 24 hours of implantation. Results from ischemia-reperfusion injury, prolonged ischemic time, donor-recipient size mismatch, and pre-existing donor cardiac dysfunction. The leading cause of early post-transplant death, often requiring ECMO.	Occurs in 10-20% of cardiac transplant recipients; severe PGD (requiring ECMO) in 3-10%. Mortality from severe PGD: 30-50%. Risk factors: prolonged ischemic time (>4-5 hours), donor cardiac dysfunction, recipient pre-transplant ECMO, pulmonary hypertension in recipient. US: ~100–200 new US pediatric PGD cases/yr Global: ~3,900–7,800 global Source: PGD in 10–20% of cardiac transplant recipients; severe PGD requiring ECMO in ~3–5%	Clinical presentation: failure to wean from cardiopulmonary bypass, or acute severe hemodynamic deterioration within first 24 hours. Echo: severe biventricular dysfunction (EF <40%). Right heart catheterization: low cardiac output, elevated filling pressures. Exclusion of: anastomotic complications, hyperacute rejection.	SOC: ECMO: primary rescue for severe PGD (VA configuration). Inotropes: epinephrine, milrinone, vasopressin. Inhaled nitric oxide for RV failure component. ECMO weaning when graft recovery sufficient (typically 48–96 hours). VAD as bridge to re-transplantation if no recovery. Prevention: minimize ischemic time, optimize donor management. Emerging: Ex vivo heart perfusion (EVHP — Transmedics OCS Heart): extends ischemic tolerance, allows real-time graft assessment, enables use of marginal donor hearts. Normothermic regional perfusion (NRP) for DCD (donation after circulatory death) hearts — expanding donor pool.
97	Cardiac Allograft Vasculopathy (CAV)	Accelerated, diffuse coronary artery disease affecting the transplanted heart, caused by chronic	Present in ~30% of pediatric recipients by 5 years and ~50% by 10 years	Annual coronary assessment: cardiac catheterization with IVUS (gold standard -- detects	SOC: Statin therapy from day 1 post-transplant (reduces CAV progression). Everolimus or sirolimus (mTOR inhibitors): most evidence for slowing CAV — switch

		immune-mediated vascular injury. Unlike native coronary artery disease, CAV affects the entire coronary tree diffusely and concentrically. The leading cause of death in pediatric cardiac transplant recipients beyond the first year.	on IVUS. Annual incidence of severe CAV: ~2-3%/year. CAV substantially limits long-term transplant survival. US: ~750–1,000 US pediatric CAV cases; ~3,000–5,000 US total (all ages) Global: ~29,000–48,000 global Source: CAV present in ~30% of pediatric recipients by 5 years and ~50% by 10 years on IVUS	CAV before angiographic changes). Coronary angiography (underestimates disease extent). Dobutamine stress echo (functional ischemia assessment). Surveillance: annually from year 1 post-transplant. Biomarkers: donor-specific antibodies (DSA) -- major risk factor for CAV.	from azathioprine in patients with early CAV. Optimize immunosuppression. Treat DSA: plasmapheresis + IVIG + rituximab. PCI for focal lesions. Retransplantation for advanced CAV. Emerging: Belatacept (CTLA4-Ig) as calcineurin-sparing strategy to reduce nephrotoxicity while maintaining immunosuppression. OCT (optical coherence tomography) for early intimal thickness quantification and CAV staging. dd-cfDNA as liquid biopsy for subclinical rejection monitoring.
98	Post-Transplant Lymphoproliferative Disorder (PTLD)	EBV-driven lymphoid proliferation occurring in immunosuppressed cardiac transplant recipients, ranging from benign polyclonal hyperplasia to aggressive monoclonal lymphoma. Pediatric recipients at highest risk because most are EBV-naïve at transplantation.	PTLD incidence in pediatric cardiac transplant recipients: 5-15%; EBV-naïve recipients have highest risk (10-20%). Peak incidence in first 2 years post-transplant. Most common malignancy post-cardiac transplant. Mortality from PTLD: 20-40%. US: ~100–200 new US pediatric PTLD cases/yr Global: ~3,900–7,800 global Source: PTLD incidence ~5–15% in pediatric cardiac transplant	EBV viral load monitoring: quantitative PCR for EBV DNA (surveillance every 1-3 months in first 2 years). Rising EBV viral load precedes PTLD development. CT chest/abdomen/pelvis: lymphadenopathy, mass lesions. PET scan for staging. Tissue biopsy: definitive diagnosis and WHO classification.	SOC: Immunosuppression reduction (first-line — induces remission in ~30–50% of early polyclonal PTLD). Rituximab (anti-CD20): first-line for CD20+ B-cell PTLD (>90% of pediatric PTLD). CHOP-based chemotherapy for refractory or aggressive monoclonal PTLD. Ganciclovir/valganciclovir prophylaxis in EBV-naïve recipients. Emerging: EBV-specific cytotoxic T-lymphocyte (CTL) adoptive immunotherapy — available at specialized centers; highly effective for EBV+ PTLD. PD-1 checkpoint inhibitors (pembrolizumab) for refractory PTLD (caution: rejection risk — use with augmented immunosuppression).

			recipients; EBV-naive recipients highest risk		
INTERVENTIONAL CARDIOLOGY — CATHETERIZATION LAB CONDITIONS					
#	CONDITION	DESCRIPTION	FREQUENCY	DIAGNOSIS	TREATMENT
99	Branch Pulmonary Artery Stenosis	Narrowing of the left and/or right pulmonary artery branches and their segmental/subsegmental divisions. Occurs as a congenital lesion or, more commonly, post-operatively following repair of TOF, truncus arteriosus, TAPVR, PA-VSD, and after Glenn/Fontan procedures. Causes elevated RV pressure, RV hypertrophy and failure, and reduced pulmonary blood flow to affected lung segments.	Post-operative PA stenosis is virtually universal to some degree after repairs involving pulmonary arterial reconstruction. Clinically significant branch PA stenosis requiring intervention: ~20-30% of TOF repairs, ~30-40% of truncus arteriosus repairs, and essentially all patients with PA-VSD/MAPCAs. Williams syndrome causes severe diffuse peripheral PA stenosis in ~75-90% of patients. US: ~5,000–15,000 US total (all degrees); ~1,000–3,000 clinically significant Global: ~195,000–585,000 global Source: Post-operative PA stenosis virtually universal after PA-related CHD repairs; clinically significant in ~5–20%	Echo with Doppler (RV pressure estimation via TR jet velocity). Cardiac MRI: gold standard for bilateral branch PA assessment and differential lung blood flow quantification. Cardiac catheterization + angiography: pressure gradient measurement (>20 mmHg generally warrants intervention). CT angiography for segmental/subsegmental anatomy. Radionuclide perfusion scan (V/Q) for differential lung flow.	SOC: Transcatheter balloon pulmonary arterioplasty (initial treatment for discrete stenoses). PA stenting with bare metal stents (allows re-dilation as child grows — preferred for most significant lesions). Cutting balloon for fibrotic or elastic recoil lesions. Surgical arterioplasty for bifurcation stenoses not catheter-accessible. Emerging: Drug-eluting stents (sirolimus-eluting) for PA stenosis — under investigation to reduce restenosis. High-pressure compliant balloons (ATLAS balloon) for resistant lesions. Drug-eluting balloons (paclitaxel-coated) for restenotic lesions. Biodegradable vascular scaffolds in early research.
100	RV-PA Conduit Failure	Stenosis and/or regurgitation of right	Conduit failure requiring	Echo: Doppler-estimated conduit gradient, degree	SOC: TPVR: preferred for suitable conduit anatomy. Melody valve (conduits 14–28

		ventricle-to-pulmonary artery conduits placed during initial surgical repair of TOF, pulmonary atresia with VSD, truncus arteriosus, and Ross procedure. Conduits are inevitably too small at initial implantation and undergo progressive calcification and stenosis with time. Each conduit historically required open-heart surgical replacement every 8-12 years. Transcatheter pulmonary valve replacement (TPVR) has fundamentally changed this management.	intervention occurs in virtually all conduit recipients over time. Estimated 250,000-300,000 patients in the US live with an RV-PA conduit. Melody Valve (Medtronic, FDA 2010) and SAPIEN XT/3 (Edwards, FDA 2016) have collectively performed >50,000 TPVR procedures worldwide. US: ~2,000–5,000 new US conduit interventions/yr Global: ~78,000–195,000 global interventions/yr Source: Conduit failure requiring intervention in virtually all recipients over time; median conduit longevity ~10–15 yrs	of PR, RV size and function. Cardiac MRI: gold standard for PR quantification (RV EDVi >150 mL/m² or RVEF <47% are standard intervention thresholds). CT angiography for conduit anatomy, coronary artery proximity, conduit diameter measurement (essential for valve sizing). Cardiac catheterization: conduit hemodynamics, balloon conduit sizing, coronary compression test before TPVR (mandatory).	mm). SAPIEN 3 (conduits/native RVOT 19–29 mm). Harmony TPV (native RVOT, FDA 2021). Coronary compression test mandatory before implantation. Surgical conduit replacement when conduit too large, calcified, or TPVR coronary compression risk prohibitive. Emerging:Harmony TPV22 (next-generation, broader sizing) for dilated native RVOT. COMPASSION trials expanding TPVR to non-circular/dilated native RVOT. Tissue-engineered conduits (decellularized, bioresorbable) to reduce reintervention burden.
101	Acquired Pulmonary Vein Stenosis	Progressive stenosis of one or more pulmonary veins occurring after surgical repair of TAPVR, or arising de novo in premature infants without prior surgery. Distinct from congenital pulmonary vein disease (TAPVR, #17). The post-surgical form is one of the most challenging conditions in pediatric interventional cardiology: stenosis is	Clinically significant acquired PV stenosis after TAPVR repair: ~10-20% of patients. Onset typically within 6-24 months of surgical repair. Bilateral or multivessel involvement in ~50% of cases. Mortality without successful	Echo: reduced or absent flow in affected pulmonary veins; elevated left atrial pressure. Cardiac MRI or CT angiography: delineates PV anatomy, degree of stenosis, and multivessel disease. Cardiac catheterization + selective PV angiography: gold standard for stenosis severity and anatomy. Lung perfusion scan (V/Q): quantifies	SOC:Transcatheter balloon PV angioplasty. Drug-eluting balloon (paclitaxel-coated): improving short-term results over plain balloon. Sutureless PV liberation surgery: creates wider, pericardium-lined pathway. Lung transplantation for severe bilateral disease not amenable to other interventions. Emerging:Sirolimus (systemic mTOR inhibitor): active trials at Boston Children's and Nationwide Children's — anti-proliferative effect on myofibroblast-driven stenosis; most promising emerging therapy. Drug-eluting stents (sirolimus-eluting) for in-

		highly aggressive in young infants, often multivessel, and recurs rapidly (50-70% restenosis within months). Carries very high mortality without successful intervention.	intervention: >50% within 2 years of diagnosis. US: ~200–500 US total clinically significant acquired PVS cases Global: ~7,800–19,500 global Source: Clinically significant acquired PV stenosis after TAPVR repair in ~10–20% of patients	differential blood flow. Wedge angiography technique for distal stenosis visualization.	stent restenosis. Imatinib + bevacizumab antifibrotic combination (investigational).
102	Coronary Artery Fistula	Congenital anomalous communication between a coronary artery and a cardiac chamber (coronary-cameral fistula: most commonly to RV or RA) or a vascular structure (coronary-arteriovenous fistula: most commonly to pulmonary artery). Causes volume loading of the drainage chamber, myocardial ischemia (coronary steal), and risk of bacterial endocarditis.	Prevalence: ~0.1–0.2% of CHD. Incidental discovery accounts for the majority. Symptomatic large fistulas: heart failure (volume loading), myocardial ischemia, bacterial endocarditis, spontaneous rupture (rare). US: ~360–720 US total (0.1–0.2% of CHD) Global: ~14,000–28,000 global Source: Coronary artery fistula ~0.1–0.2% of CHD; majority discovered incidentally	Echo: turbulent flow jet entering cardiac chamber or pulmonary artery; dilated coronary artery. CT coronary angiography: gold standard for anatomy, fistula tract, drainage site. Cardiac catheterization + selective coronary angiography: fistula tract anatomy, Qp/Qs measurement, branch vessel assessment before embolization. Stress testing for ischemia (large fistulas with steal).	SOC: Transcatheter embolization: preferred for suitable anatomy — coil embolization (detachable/pushable) for small-to-medium fistulas; Amplatzer Vascular Plug for large-diameter fistulas. Technical priority: preserve feeding coronary branches. Surgical ligation for anatomy not amenable to catheter closure. Observation for small asymptomatic fistulas. SBE prophylaxis post-closure. Emerging: Covered stent exclusion for large fistulas as an alternative to embolization. CT coronary angiography for pre-procedural mapping and post-closure surveillance. Multimodality imaging (FFR, IVUS) to assess hemodynamic significance of small fistulas.
103	Pulmonary Arteriovenous Malformations (Post-Glenn)	Abnormal direct connections between pulmonary arteries and pulmonary veins in the lung parenchyma, bypassing the gas	Clinically significant PAVMs after bidirectional Glenn anastomosis: ~10-15% of patients. Develop in virtually	Clinical presentation: progressive cyanosis after initially adequate Glenn palliation, without echocardiographic evidence of shunting.	SOC: Fontan completion — definitive treatment; restoration of hepatic venous blood flow reverses PAVMs in most cases (regression typically 3–12 months post-Fontan). Transcatheter embolization for large discrete accessible PAVMs

		exchange capillary bed. In CHD, PAVMs develop characteristically after the bidirectional Glenn procedure as a consequence of the absence of hepatic venous blood flow from the pulmonary circulation. The hepatic factor (a protein in hepatic venous effluent) normally prevents PAVM formation; its exclusion after Glenn allows PAVMs to develop. Causes progressive cyanosis and limits Fontan candidacy until hepatic flow is restored.	all patients with interrupted IVC/azygous continuation who undergo a classic Glenn. PAVMs typically regress after Fontan completion when hepatic venous blood is delivered to the pulmonary circulation. US: ~1,000–2,500 US Glenn patients with PAVMs Global: ~39,000–97,500 global Source: Clinically significant PAVMs in ~10–15% of patients after bidirectional Glenn anastomosis	Contrast echo: microbubbles appear in left heart after peripheral venous injection without passing through the heart (pulmonary A-V passage) -- diagnostic. CT pulmonary angiography: visualizes PAVM locations. Cardiac MRI: differential lung blood flow. Cardiac catheterization: pulmonary vein saturation step-downs; selective pulmonary angiography.	(temporizing). Hepatic vein inclusion in Fontan pathway is critical for anatomical variants with hepatic vein exclusion. Emerging: Hepatic vein flow redistribution: catheter-based selective hepatic venous redirection as bridge to Fontan completion. Understanding of hepatic factor biology (identity of anti-PAVM hepatic factor) under active investigation — may lead to pharmaceutical approaches.
104	Kawasaki Giant Coronary Aneurysm with Stenosis	Distinct, late-phase complication of Kawasaki disease characterized by progressive stenosis within or adjacent to giant coronary artery aneurysms (Z-score ≥ 10 or absolute diameter ≥ 8 mm). Stenosis results from myofibroblast proliferation and organized thrombus within the aneurysm. Causes myocardial ischemia, MI, and sudden cardiac death. Distinct from acute Kawasaki disease (#79): requires interventional coronary management rather than anti-inflammatory treatment.	Giant coronary aneurysms (Z-score ≥ 10) develop in ~1% of untreated and ~0.1% of treated Kawasaki patients. Stenosis within giant aneurysms: clinically significant in ~50% of giant aneurysm patients by 10-20 years of follow-up. MI from Kawasaki coronary disease: 5-10% of giant aneurysm patients. US: ~200–1,000 new US Kawasaki	Coronary angiography (catheterization): gold standard for anatomy and stenosis severity. CT coronary angiography: non-invasive surveillance (annual or biannual). Stress imaging (nuclear perfusion, stress MRI, dobutamine stress echo): ischemia assessment. IVUS and OCT: assess aneurysm wall, thrombus, and stenosis anatomy -- critical for intervention planning. Echo: aneurysm size follow-up, LV wall motion.	SOC: Lifelong anticoagulation: warfarin + low-dose aspirin for giant aneurysms (z-score ≥ 10). PCI: balloon angioplasty $\pm$ rotational atherectomy (calcified lesions) + drug-eluting stents preferred. CABG: for multivessel or PCI-unsuitable disease — internal mammary artery conduits preferred (avoid saphenous vein in young patients). Thrombolytics or catheter-directed aspiration for acute coronary occlusion. Emerging: Orbital atherectomy for heavily calcified KD coronary lesions. IVUS and OCT for lesion assessment and stent optimization. Direct oral anticoagulants (DOACs — rivaroxaban, apixaban) as alternatives to warfarin for anticoagulation — comparative trials ongoing. Fractional flow reserve (FFR) to guide PCI indications.

			giant aneurysm cases/yr; ~8,000–20,000 US with any CA involvement Global: ~7,800–39,000 new global giant aneurysm cases/yr Source: Giant coronary aneurysms (Z-score ≥10) develop in ~1% of untreated and ~0.1% of treated Kawasaki patients		
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Pediatric Cardiac Conditions: Hereditary Predisposition Syndromes

41 conditions with meaningful germline components, drawn from the 104-condition Pediatric Cardiac Atlas

Tier Key: 🧬🧬🧬 Defining (condition IS a germline syndrome) 🧬🧬 Dominant (germline mutation in >25% of cases; family screening standard of care) 🧬🧬 Associated (strong syndromic link; not universal)

These 41 conditions represent the highest-leverage targets for federated AI surveillance across children's hospitals. Identifying germline carriers before cardiac disease presents -- and surfacing at-risk family members from population-level genotype and phenotype data within hospital firewalls -- can meaningfully shift outcomes in ways that treating the index patient alone cannot. The Tier 3 conditions in particular align with a disease-precise AI agent framework: patients with Marfan syndrome, LQTS, HCM, ARVC, and 22q11.2 deletion require lifelong, protocol-driven surveillance that federated models can automate at scale across distributed hospital networks.

🧬🧬🧬 TIER 3 — DEFINING: The cardiac condition IS a germline syndrome or near-universally caused by an identified genetic mutation						
TIER	#	CONDITION	SYNDROME / GENE(s)	INHERITANCE	CLINICAL NOTES	CARDIAC KEY POINTS
🧬🧬🧬	41	Duchenne MD Cardiomyopathy	DMD gene (Xp21); dystrophin protein absent	X-linked recessive	Affects ~1 in 3,500-5,000 male births. Progressive muscle weakness and DCM. Cardiomyopathy present in virtually all patients by age 18; now the leading cause of DMD death. Female carriers: ~10-20% develop DCM.	Annual echo + ECG from age 10. Cardiac MRI with LGE for early fibrosis. ACE inhibitors (cardioprotective even pre-dysfunction). Beta blockers. Steroids. Gene therapy (Elevidys/delandistrogene moxeparovec, Sarepta, FDA 2023/2024 for ambulatory patients age 4+). Genetic counseling: female carriers need cardiac screening.
🧬🧬🧬	42	Emery-Dreifuss MD / LMNA Cardiomyopathy	LMNA (lamin A/C) or EMD (emerin) mutations	AD (LMNA) or X-linked (EMD)	Classic triad: early joint contractures, slowly progressive muscular dystrophy, and life-threatening cardiac conduction disease progressing to DCM. Highest SCD risk of any inherited condition; >5%/year in high-risk LMNA subgroups. Cardiac disease often precedes or outpaces the muscular phenotype.	ECG: atrial standstill, progressive heart block (pathognomonic). Holter for VT. Early and aggressive ICD implantation -- mainstay of care even before significant DCM. Pacemaker for symptomatic bradycardia/heart block. Standard heart failure therapy. Cardiac transplantation for advanced disease. First-degree relatives must be screened.

	44	Pompe Disease (Glycogen Storage Type II)	GAA gene; lysosomal acid alpha-glucosidase (GAA) deficiency	Autosomal recessive	Classic infantile form: massive HCM + hypotonia + respiratory failure within first year. All forms combined ~1 in 40,000. Newborn screening now standard in most US states. Early treatment dramatically changes cardiac trajectory.	Echo: massive concentric HCM with giant QRS complexes, shortened PR interval. Early ERT initiation critical -- cardiac function often normalizes with treatment. ERT: alglucosidase alfa (Lumizyme/Myozyme, Sanofi); cipaglucosidase alfa + miglustat (Pombiliti + Opfolda, Amicus, FDA 2023) -- next-generation ERT. Respiratory support. Cardiac function often normalizes with early ERT.
	45	Danon Disease	LAMP2 gene; lysosomal-associated membrane protein 2	X-linked dominant (males severely affected; females variably affected)	Classic triad: HCM, skeletal myopathy, and intellectual disability. WPW syndrome in ~70% of males. Rapidly progressive HCM -- often requiring transplantation. Frequently misdiagnosed as sarcomeric HCM. Males typically need transplant by ~20 years; females have milder, later-onset disease.	ICD: mandatory for SCD risk (WPW + severe HCM). Cardiac transplantation -- often the only effective long-term therapy; median age at transplantation ~20 years in males. Catheter ablation for WPW accessory pathway. Gene therapy under development. Female carriers: surveillance echo and ECG every 1-2 years from diagnosis.
	46	Fabry Disease (Cardiac)	GLA gene; alpha-galactosidase A deficiency; lysosomal storage of Gb3	X-linked recessive (males severely affected; females variably affected)	~1 in 40,000 (general); possibly 1 in 3,100 including late-onset variants. Cardiac manifestations: progressive HCM (concentric pattern), arrhythmia, heart failure. Late-onset cardiac variant (IVS3-10G>A mutation) presents with isolated cardiac disease in older males. Newborn screening available in some states.	ERT: agalsidase alfa (Replagal, EU) or agalsidase beta (Fabrazyme, Sanofi, US). Migalastat (Galafold, Amicus, oral chaperone for amenable mutations). Cardiac MRI with LGE: characteristic mid-wall inferolateral LGE pattern (pathognomonic of Fabry cardiac involvement). Serial echo every 1-2 years for HCM monitoring. Treat cardiac complications (arrhythmia, anticoagulation). Female carriers: can develop significant cardiac disease -- require regular screening.
	47	Friedreich's Ataxia (Cardiac)	FXN gene (frataxin); GAA trinucleotide repeat expansion; mitochondrial iron accumulation	Autosomal recessive	~1 in 50,000; most common inherited ataxia. Cardiac involvement (HCM + arrhythmia) present in >90% of patients. Progressive neurological disease (ataxia, dysarthria, dysphagia) accompanies cardiac disease. Cardiac disease is the leading cause of death. GAA repeat length correlates with disease severity.	Omaveloxolone (Skyclarys, FDA 2023) -- first disease-modifying therapy (age >=16); reduces neurological progression and has some cardiac benefit. Beta blockers for HCM symptoms. ACE inhibitors for heart failure. ICD consideration for arrhythmia risk. Annual echo + ECG + Holter. Unlike sarcomeric HCM, HOCM-directed therapies (myectomy, septal ablation) are rarely appropriate.
	48	Barth Syndrome (Cardiac)	TAZ gene (tafazzin); abnormal cardiolipin metabolism in inner mitochondrial membrane	X-linked recessive	~1 in 300,000-400,000 male births; significantly underdiagnosed. Classic presentation: DCM (and/or LVNC), skeletal myopathy, neutropenia, and growth retardation in infant males. Urine 3-methylglutaconic acid elevation is a biochemical hallmark.	Echo (DCM; occasionally non-compaction). Standard heart failure therapy. Nutritional support. ICD for ventricular arrhythmia. Cardiac transplantation for refractory cases. Granulocyte-stimulating factor (G-CSF) for severe neutropenia. Genetic counseling: carrier females are unaffected but must be identified for family planning.
	49	Mucopolysaccharidoses (Cardiac)	MPS I/Hurler: IDUA; MPS II/Hunter: IDS; MPS IVA/Morquio: GALNS; MPS VI: ARSB; lysosomal enzyme genes	AR (MPS I, IVA, VI); X-linked (MPS II)	Combined MPS incidence ~1 in 25,000-30,000. Progressive glycosaminoglycan accumulation in cardiac valves, coronary arteries, and myocardium. Cardiac involvement in >90%. Progressive valvular disease (predominantly mitral and aortic) is the dominant cardiac feature. Difficult airway: major anesthetic consideration in all MPS patients.	ERT: laronidase (MPS I/Aldurazyme, Sanofi), idursulfase (MPS II/Elaprase, Takeda), galsulfase (MPS VI/Naglazyme), elosulfase alfa (MPS IVA/Vimizim). HSCT for MPS I Hurler before age 2.5 years -- can attenuate cardiac progression if performed early. Serial echo every 1-2 years for valve disease tracking. Surgical valve repair or replacement for advanced valve disease. Difficult airway management requires specialized anesthetic team.
	50	Mitochondrial Cardiomyopathy	mtDNA mutations (MELAS: MT-TL1; KSS: mtDNA deletion; Leigh: MT-ATP6) or nuclear genes (POLG, TWNK); failure of oxidative phosphorylation	Maternal (mtDNA mutations); AR or AD (nuclear gene mutations)	Cardiac involvement in ~20-40% of all mitochondrial disease patients. Major syndromes: MELAS, Kearns-Sayre syndrome (complete heart block + HCM), Leigh syndrome (severe infantile HCM), MERRF. Unlike sarcomeric HCM, myectomy and septal reduction are rarely appropriate. Cardiac	Pacemaker: mandatory for complete heart block in KSS -- sudden death from heart block is a leading cause of KSS mortality; cannot be deferred. ICD for sustained VT/VF. Standard heart failure therapy. CoQ10 supplementation (widely used; evidence limited). Riboflavin for Complex I deficiency subtypes. Avoid valproate (inhibits mitochondrial beta-oxidation).

					transplantation controversial because multisystem disease continues after transplant.	Cardiac transplantation only after careful multidisciplinary review. Genetic counseling critical: maternal inheritance pattern (mtDNA mutations) vs Mendelian (nuclear gene mutations) has critical implications for family screening.
	51	Marfan Syndrome (Cardiac)	FBN1 gene (fibrillin-1); TGF-beta signaling dysregulation	Autosomal dominant (25% de novo)	~1 in 5,000. Cardinal cardiac features: progressive aortic root dilation (risk of Type A dissection and rupture), aortic valve regurgitation, and mitral valve prolapse. Aortic dissection risk ~5-10%/year once root >5cm. Ghent criteria 2010 for diagnosis. Family screening mandatory for first-degree relatives.	Losartan or atenolol -- reduce aortic root growth rate. Prophylactic aortic root replacement when root >4.5-5.0cm (lower threshold in young patients, athletes, rapid progression, or family history of dissection). Valve-sparing root repair preferred in young patients (David or Yacoub procedure). Exercise restriction. Annual echocardiographic surveillance; annual MRI/CT to image entire aorta.
	52	Loeys-Dietz Syndrome	TGFBR1, TGFBR2, SMAD3, TGFB2, or TGFB3 mutations; TGF-beta pathway	Autosomal dominant	~1 in 50,000-100,000. More aggressive aortic dilation and dissection at SMALLER diameters than Marfan syndrome -- dissection can occur at 4.0cm or even smaller. TGFBR2 mutations carry highest risk. Affects entire arterial tree: cerebrovascular, visceral, and spinal arteries also at risk.	Losartan -- first-line therapy (more evidence than in Marfan). Lower threshold for prophylactic aortic surgery (~4.0-4.5cm vs. 5.0cm in Marfan). Pan-arterial surveillance from head to pelvis: annual full-body MRI/CT. Exercise restriction. Genetic testing of all first-degree relatives. Pregnancy carries high risk: managed at specialized centers with lower surgical thresholds.
	53	Turner Syndrome (Cardiac)	45,X karyotype or mosaicism; haploinsufficiency of X-chromosome genes	Chromosomal aneuploidy (not familial)	~1 in 2,000-2,500 live female births. Cardiac findings in ~30-50%: BAV (~16%), coarctation (~10%), aortic dilation. Dissection risk 6-10x higher than general population despite relatively modest aortic dilation. Cardiovascular disease is the leading cause of premature mortality in Turner syndrome. Mosaic Turner: milder phenotype but same cardiac surveillance required.	Echocardiography at diagnosis and every 3-5 years (or annually if aortic dilation). MRI/CT for full aortic assessment. Aortic size index (ASI) -- dissection risk increases significantly at ASI >2.5 cm/m2. Hormone replacement therapy (estrogen) -- does not increase aortic risk. Antihypertensive therapy. Prophylactic aortic surgery at lower thresholds than general population. Coarctation repair per standard protocols. Pregnancy contraindicated if ASI >2.5 cm/m2.
	54	Vascular Ehlers-Danlos Syndrome (EDS Type IV)	COL3A1 gene; type III procollagen deficiency	Autosomal dominant (50% de novo)	~1 in 50,000-200,000. Most life-threatening EDS subtype. Unlike Marfan/Loeys-Dietz, the aortic root is often NORMAL -- danger lies in spontaneous rupture of medium-sized arteries WITHOUT preceding aneurysm. Arterial tissue is extraordinarily fragile; sutures tear through. First major complication by age 40 in ~80%; median survival ~48 years. Operating on an undiagnosed vascular EDS patient is a recognized cause of intraoperative catastrophe.	Celiprolol (beta blocker with vasodilatory properties): only agent with randomized evidence (BBEST trial) for reducing arterial event rates. Conservative management strongly preferred. Surgical intervention for life-threatening rupture only: modified technique mandatory -- pledgeted sutures, Teflon felt reinforcement, minimal tissue handling; endovascular preferred over open. Avoid invasive arterial procedures (arterial puncture can cause uncontrollable hemorrhage). MRI/MRA preferred over invasive angiography for surveillance. Pre-operative screening: all patients undergoing cardiac/vascular surgery should be screened for connective tissue disorders.
	58	Homozygous Familial Hypercholesterolemia (HoFH)	LDLR (both alleles), or APOB, or PCSK9 gain-of-function; absent LDL receptor function	Autosomal recessive (HoFH); biallelic mutations	1 in 300,000-1,000,000. Hundreds to a few thousand US patients. LDL 400-1,000+ mg/dL from birth. Aggressive atherosclerosis in childhood; MI before age 20 without treatment. Cutaneous and tendon xanthomas typical. Aortic stenosis from lipid infiltration of the aortic valve.	High-intensity statins + ezetimibe + PCSK9 inhibitors (less effective than in heterozygous FH due to absent LDL receptors). Evincumab (Evinacea, Regeneron, FDA 2021) -- ANGPTL3 inhibitor; reduces LDL ~49% independent of LDL receptors. LDL apheresis every 1-2 weeks (standard of care). Lomitapide (MTP inhibitor). Coronary surveillance: CT coronary angiography, stress imaging. Aortic valve surveillance. Early treatment is imperative to prevent irreversible coronary disease.
	59	Long QT Syndrome (LQTS)	KCNQ1 (LQT1), KCNH2 (LQT2), SCN5A (LQT3); >17 subtypes; ion channel mutations affecting cardiac	Autosomal dominant (Romano-Ward); AR (Jervell-Lange-Nielsen syndrome)	~1 in 2,000. ~3,500 sudden deaths annually in the US. >90% of genotyped cases are LQT1 (swimming/exercise trigger) or LQT2 (sudden sounds trigger). Jervell-Lange-Nielsen	Beta blockers (first-line; nadolol or propranolol preferred); very effective in LQT1 and LQT2; mexiletine add-on for LQT3. ICD for high-risk patients (LQT3, prior cardiac arrest, breakthrough events on beta

			repolarization	with deafness)	(biallelic KCNQ1) is the most severe form with congenital deafness and highest SCD risk. Acquired LQTS from QT-prolonging drugs is far more common than congenital forms.	blockers). Avoidance of QT-prolonging medications (www.crediblemeds.org). Left cardiac sympathetic denervation (LCSD) for refractory cases -- highly effective, especially in LQT1 and LQT2. Activity restriction by genotype: LQT1 -- avoid competitive swimming; LQT2 -- avoid sudden loud alarms. Family screening mandatory: 12-lead ECG + genetic testing for all first-degree relatives.
🧬🧬🧬	60	CPVT (Catecholaminergic Polymorphic VT)	RYR2 (ryanodine receptor 2) ~60% of cases; CASQ2 (calsequestrin 2) ~5%; abnormal intracellular calcium handling	AD (RYR2); AR (CASQ2)	~1 in 10,000. Accounts for ~15% of autopsy-negative sudden cardiac death in the young. Structurally normal heart with normal resting ECG. SCD is often the FIRST manifestation -- before any diagnosis. Characteristic bidirectional VT (beat-to-beat alternating QRS axis) or polymorphic VT during exercise or emotional stress is diagnostic.	Beta blockers (nadolol preferred over metoprolol/atenolol due to efficacy data). Flecainide (add-on, reduces VT burden significantly -- now standard of care in symptomatic CPVT). ICD for survivors of cardiac arrest or breakthrough VT events. Left cardiac sympathetic denervation (LCSD) as adjunctive therapy -- effective as add-on. Strict exercise restriction -- mandatory. Family screening: exercise stress test + genetic testing for all first-degree relatives. Avoid sympathomimetic medications.
🧬🧬🧬	61	Brugada Syndrome	SCN5A (sodium channel Nav1.5) mutations in ~20-30%; many cases remain genotype-negative	Autosomal dominant (incomplete penetrance)	~1 in 2,000 globally; higher prevalence in Southeast Asia. Male predominance 8-10:1. VF risk during rest or sleep (vagal state). Fever dramatically increases risk by accentuating ST changes -- fever management is critical and life-saving. Many patients are asymptomatic until VF or aborted cardiac arrest.	ICD -- only proven therapy for symptomatic patients (prior VF or syncope presumed to be arrhythmic). Quinidine (Ito current blocker) for VT storms, ICD-eligible patients, or high-risk asymptomatic patients with inducible VF. Aggressive fever management (antipyretics at first sign of fever; ER visit if >38.5°C). Avoidance of sodium channel-blocking drugs and other Brugada-aggravating drugs (www.brugadadrugs.org). S-ICD increasingly used in younger patients. EP study for risk stratification in asymptomatic patients remains controversial.
🧬🧬🧬	62	Short QT Syndrome (SQTS)	KCNQ1, KCNH2, KCNJ2 (gain-of-function potassium channel mutations -- opposite of LQTS); CACNA1C, CACNB2	Autosomal dominant	Estimated prevalence <1 in 10,000; significantly underdiagnosed. All age groups including infants (recognized cause of SIDS). SCA or VF may be the first manifestation. Critically distinct from LQTS: opposite pathophysiology; beta blockers used in LQTS are INEFFECTIVE or potentially HARMFUL in SQTS. ~70 families described in literature as of 2023.	ICD: mainstay for symptomatic patients (survivors of VF/SCA). T-wave oversensing is a known technical problem with ICD in SQTS due to tall, peaked T-waves -- careful programming required. Quinidine or hydroquinidine: reduces VF inducibility and prolongs QT; used in patients not eligible for ICD (infants) or as adjunct. Beta blockers: DO NOT USE as primary therapy (ineffective and potentially harmful). Family screening with 12-lead ECG. Activity restriction pending ICD placement.
🧬🧬🧬	63	Noonan Syndrome (Cardiac)	PTPN11 (~50%), RAF1, SOS1, RIT1, HRAS, KRAS, BRAF; RASopathy (RAS-MAPK pathway dysregulation)	Autosomal dominant (50% de novo)	~1 in 1,000-2,500 live births. Most common syndromic cause of CHD after Down syndrome. Cardinal cardiac features: pulmonary stenosis (50-60%), HCM (20-30%), ASD. HCM in Noonan is often refractory and may progress rapidly in infancy -- distinct from sarcomeric HCM and requires different management. Short stature, typical facies, cryptorchidism, lymphedema.	Pulmonary stenosis: balloon valvuloplasty (dysplastic valves have lower success rates than typical PS; surgical repair needed more often). HCM: beta blockers, CCBs; trametinib (MEK inhibitor, Novartis, FDA Breakthrough Designation for HRAS-mutant Noonan HCM) -- targeted RAS-MAPK pathway therapy. Surgical myectomy for refractory LVOTO. ICD for high-risk HCM. Avoid bevacizumab and other VEGF inhibitors (risk of arteriovenous malformations in RASopathies).
🧬🧬🧬	64	Williams Syndrome / SVAS	ELN gene (elastin) deletion at 7q11.23; contiguous gene deletion syndrome (~1.5-1.8 Mb deletion)	Autosomal dominant de novo (rarely inherited)	~1 in 7,500 live births. SVAS present in ~75% and is the leading cause of morbidity. Bilateral coronary ostial involvement creates risk of sudden death during cardiac procedures (even minor ones including dental procedures under general anesthesia). Intellectual disability, hypersociability, hypercalcemia, renal artery stenosis.	Surgical SVAS repair (patch aortoplasty) for significant gradients. Doty three-patch aortoplasty preferred (see #12). Peripheral PA stenosis: transcatheter balloon dilation +/- stenting -- serial procedures required throughout childhood. CRITICAL ANESTHESIA WARNING: avoid general anesthesia without comprehensive cardiac evaluation due to SCD risk from coronary ostial involvement -- must rule out

						bilateral coronary ostial adherence pre-operatively. Lifelong cardiac surveillance. Blood pressure monitoring (bilateral arm and leg) at every visit for renal artery stenosis screening.
	55b	22q11.2 Deletion Syndrome (Cardiac)	TBX1 gene haploinsufficiency; most common microdeletion syndrome; 3 Mb deletion at 22q11.2	Autosomal dominant de novo (~90%); inherited (~10%)	~1 in 4,000 live births. Cardiac defects in 75-80%: conotruncal anomalies predominate -- Interrupted Aortic Arch (50-70%), TOF (~15%), Truncus Arteriosus (~15%). Associated immune deficiency (DiGeorge), hypocalcemia, palatal abnormalities, and psychiatric illness (schizophrenia in ~25% by adulthood). Highly variable expressivity -- same deletion causes markedly different severity within families.	Cardiac repair per specific conotruncal defect (IAA, TOF, Truncus). Calcium supplementation (neonatal hypocalcemia -- seizures if missed). Thymic transplantation or T-cell reconstitution for severe DiGeorge immune deficiency. Irradiated blood products mandated if immunocompromised (to prevent graft-versus-host disease from transfusion). Psychiatric and neurodevelopmental surveillance throughout adolescence and adulthood. Screen all first-degree relatives: FISH or microarray. Genetic counseling for affected individuals of reproductive age (50% transmission risk).
	65	Alagille Syndrome (Cardiac)	JAG1 (~94%) or NOTCH2 (~6%); Notch signaling pathway; haploinsufficiency	Autosomal dominant (50-70% de novo)	~1 in 30,000 live births. Causes bile duct paucity (cholestatic liver disease), distinctive facial features, vertebral anomalies (butterfly vertebrae), posterior embryotoxon, and cardiovascular disease. Cardiac manifestation: diffuse peripheral pulmonary arterial stenosis (bilaterally, involving segmental and subsegmental branches) in ~90% of patients. PA stenosis is the leading cause of cardiac-related mortality. Liver transplantation does NOT resolve the cardiac manifestations.	Transcatheter balloon pulmonary arterioplasty and stenting: mainstay of catheterization management throughout childhood; goal is to reduce RV pressure to sub-systemic levels and promote PA growth. Serial procedures required. Cutting balloon and stent-based approaches for fibrotic lesions. Annual cardiac MRI and echo for RV pressure tracking. Liver transplantation for biliary cirrhosis (~20-30%; does not directly treat PA stenosis). Gene therapy strategies targeting JAG1/Notch pathway: preclinical development.
	66	CHARGE Syndrome	CHD7 gene mutations (~90%); chromodomain helicase DNA binding protein 7; chromatin remodeler for neural crest cell migration	Autosomal dominant (virtually all de novo)	~1 in 8,500-12,000 live births. Cardiovascular malformations in ~75-80%: predominantly conotruncal (TOF, IAA, AVSD, VSD, PA-VSD, heterotaxy-spectrum lesions). One of the most common causes of combined deaf-blindness in children. Choanal atresia causes neonatal respiratory distress (obligate nasal breathers). Semicircular canal aplasia causes profound vestibular dysfunction -- implications for sedation and post-op recovery.	Cardiac repair: standard surgical management per specific defect. Sequential surgery: choanal atresia repair and cardiac repair must be carefully prioritized -- choanal atresia may need addressing before or concurrent with cardiac surgery. Anesthetic considerations: difficult airway (choanal atresia, subglottic stenosis); no intact balance system (semicircular canal aplasia) -- prolonged sedation causes disorientation; sensorineural hearing loss affects post-op communication. Cochlear implantation for profound deafness (vestibular aplasia complicates but does not preclude implantation). Multidisciplinary management.
 TIER 2 — DOMINANT: Germline mutation identified in >25% of cases; strong causal role; genetic testing and family screening standard of care						
TIER	#	CONDITION	SYNDROME / GENE(s)	INHERITANCE	CLINICAL NOTES	CARDIAC KEY POINTS
	1	Bicuspid Aortic Valve	NOTCH1, FBN1, GATA5, and others; genetic etiology in ~25-30% of familial cases; incomplete penetrance	AD with variable penetrance; complex/multifactorial in many cases	1-2% of population; most common CHD overall. Men 2-3x more affected. ~9% of first-degree relatives of BAV probands have BAV -- the highest sibling recurrence risk of any CHD. Associated with aortic root and ascending aortic dilation independent of valve function -- the aortopathy has a genetic basis distinct from the hemodynamic effect of the valve.	Surveillance echocardiography for both valve function AND aortic dimensions. Family screening: echocardiography recommended for all first-degree relatives of BAV patients. Aortic root/ascending aorta: surgical repair or replacement when diameter >4.5cm (or >4.0cm with rapid growth, family history of dissection, or planned pregnancy). Valve repair or replacement (surgical or TAVR); Ross procedure in young patients. Annual MRI/CT if ascending aorta >4.0cm.
	12	Supravalvar Aortic Stenosis (SVAS)	ELN gene (elastin) deletion (Williams syndrome 75%) or ELN point mutations	AD (familial isolated ELN) or de novo deletion (Williams)	Williams syndrome-associated SVAS: ~75% of all SVAS cases; caused by ELN deletion 7q11.23 -- bilateral PA stenosis co-exists;	Surgical repair indicated for gradient >50 mmHg, symptoms, or LV dysfunction. Doty three-patch aortoplasty: preferred technique -- inverted Y incision

			(isolated/non-syndromic 25%); AD inheritance in familial isolated cases	syndrome)	intellectual disability; hypercalcemia; renal artery stenosis. Isolated/non-syndromic SVAS: ~25% -- ELN point mutations; no Williams features; normal neurodevelopment; excellent outcomes after repair. Familial isolated SVAS: autosomal dominant; clusters reported.	extended into all three aortic sinuses; avoids asymmetric repair of single-patch techniques. Critical intraoperative step: mandatory inspection of coronary ostia for adherence to supra-aortic ridge -- release of adherent commissures if present. Co-existing Williams bilateral PA stenosis: transcatheter PA stenting or surgical arterioplasty. Lifelong surveillance echocardiography for re-stenosis (10-20% recurrence).
	16	Interrupted Aortic Arch	TBX1 haploinsufficiency (22q11.2 deletion) in 50-70%; genetic basis for remainder unknown	AD de novo (22q11.2 deletion); remainder sporadic	~1-2 per 100,000 live births. 22q11.2 deletion present in 50-70% -- the highest 22q11.2 association of any single CHD lesion. When 22q11.2 is absent, genetic basis is often unknown. Associated with VSD in virtually all cases. Complete discontinuity of the aortic arch -- survival depends on PDA; cardiovascular collapse when ductus closes.	Prostaglandin E1 emergently on diagnosis. Surgical reconstruction connecting arch segments + VSD closure. 22q11.2 testing: mandatory in all IAA patients -- guides immune management, calcium supplementation, transfusion of irradiated blood products, and family counseling. Screen all first-degree relatives of 22q11.2-positive patients. 22q11.2-positive patients require psychiatric and neurodevelopmental surveillance into adulthood.
	36	Hypertrophic Cardiomyopathy (HCM)	MYBPC3 (~40%), MYH7 (~35%), TNNT2, TNNI3, TPM1 and others; sarcomere gene mutations; ~50% of HCM patients have identifiable pathogenic variant	Autosomal dominant (incomplete penetrance and variable expressivity)	~1 in 500 individuals (~700,000 Americans). Leading cause of SCD in athletes under 35 (~36% of events). ~50% of HCM patients have identifiable pathogenic sarcomere variant; ~50% are genotype-negative (possible polygenic or unknown etiology). Phenotype can be absent in mutation carriers until adulthood. Noonan syndrome (PTPN11) causes a phenotypically similar but mechanistically distinct HCM (see #63).	Beta blockers, calcium channel blockers. Mavacamten (Camzyos, FDA 2022) -- myosin inhibitor; first disease-specific therapy for obstructive HCM. Aficamten (Myqorzo, FDA 2025). Septal myectomy or alcohol septal ablation for refractory LVOTO. ICD for high-risk patients. Family screening: genetic testing of first-degree relatives when pathogenic variant identified; echocardiography screening for all first-degree relatives regardless of genotype (every 5 years in adults; every 1-2 years in adolescents). Sport restriction counseling.
	37	Dilated Cardiomyopathy (DCM)	LMNA (5-10%), TTN (titin, 20-25% of familial DCM), SCN5A, PLN, RBM20, FLNC and >50 additional genes; 20-35% of pediatric DCM is genetic	AD most common; AR, X-linked, and mitochondrial inheritance also documented	0.57-1.13 per 100,000 children/year. Up to 50-60% idiopathic (unknown etiology); 20-35% genetic. LMNA-DCM carries particularly high SCD risk (conduction disease + VT) even at relatively preserved LVEF -- early ICD threshold. TTN truncating variants: most common genetic cause of adult DCM but challenging to interpret in pediatric population (high prevalence of benign TTN variants).	ACE inhibitors, beta blockers, diuretics. CRT for dyssynchrony. ICD. LMNA-DCM: early ICD implantation (even before LVEF <35%) due to high sudden death risk from conduction disease. Cardiac transplantation; VAD (Berlin Heart EXCOR for <15 kg; continuous-flow device for larger patients) as bridge to transplantation. Family screening: genetic testing + echocardiography in first-degree relatives of genetic DCM patients. Cascade screening identifies pre-symptomatic family members who can receive prophylactic ACE inhibitors.
	39	ARVC (Arrhythmogenic Right Ventricular Cardiomyopathy)	PKP2 (plakophilin-2, ~40% of familial ARVC), DSP (desmoplakin), DSG2 (desmoglein-2), DSC2, JUP; desmosomal gene mutations; ~50% of cases have identifiable pathogenic variant	Autosomal dominant (incomplete penetrance and variable expressivity)	~1 in 2,000-5,000. Accounts for ~11-17% of SCD in athletes under 35. Exercise dramatically accelerates disease progression and penetrance -- the single most important modifiable factor. PKP2 mutations in ~40% of familial cases. Strong evidence that competitive athletics accelerates fibro-fatty replacement and increases SCD risk in ARVC gene carriers (even before phenotypic expression).	Exercise restriction -- CRITICAL; should be implemented immediately upon genetic diagnosis even before phenotypic expression. ICD for sustained VT, VF, or syncope. Catheter ablation for refractory VT (epicardial approach often required -- ARVC VT circuits are epicardial). Beta blockers. Heart transplantation for advanced biventricular failure. Family screening: genetic testing + cardiac MRI + ECG + Holter + exercise stress test in all first-degree relatives of ARVC patients. Athletes with pathogenic desmosomal variants should be disqualified from competitive sports.
	40	Left Ventricular Non-Compaction (LVNC)	MYH7, MYBPC3, LMNA, TAZ (tafazzin), SCN5A, LDB3, HCN4 and others; ~40% of pediatric LVNC has identifiable genetic cause	AD, X-linked (TAZ), or AR depending on gene; de novo mutations common	Estimated 0.05-0.24% in echocardiographic studies; ~10% of pediatric cardiomyopathies. Overlaps with HCM (MYH7/MYBPC3) and DCM (LMNA/SCN5A). TAZ (Barth syndrome) causes LVNC in infant males -- distinct presentation with skeletal myopathy and neutropenia. HCN4 mutations: LVNC with	Standard heart failure therapy. Anticoagulation for LVEF <35% or documented thrombus. ICD for high-risk arrhythmia (particularly LMNA-associated LVNC). Cardiac transplantation for refractory heart failure. Family screening: echocardiography + genetic testing in first-degree relatives when pathogenic variant identified. Barth syndrome (TAZ): requires nutritional

					bradycardia and atrial fibrillation. Noonan syndrome: LVNC can co-exist with HCM.	support, G-CSF for neutropenia, and early heart failure management.
	56	Pulmonary Arterial Hypertension (PAH)	BMPR2 (bone morphogenetic protein receptor 2) mutations in ~80% of familial PAH and ~20% of idiopathic PAH; ACVRL1, ENG, KCNK3, ATP13A3, AQP1 and others	AD (incomplete penetrance ~20-30%)	Idiopathic PAH: 15-50 per million. Familial PAH (FPAH): ~6% of all PAH patients; BMPR2 mutation in ~80%. BMPR2 mutation carriers have ~20-30% lifetime risk of developing PAH -- incomplete penetrance; modifiers unknown. Female sex increases penetrance. PAH-CHD much more common in pediatric cardiac populations (Eisenmenger physiology, repaired defects with residual PH).	Prostanoids, ERA (ambrisentan, bosentan), PDE-5 inhibitors (sildenafil, tadalafil). Sotatercept (Winrevair, Merck, 2024) -- novel activin-pathway therapy; targets BMPR2 signaling pathway -- biologically rationale for familial PAH with BMPR2 mutations. Lung transplantation for end-stage disease. Genetic testing: offered to all PAH patients; BMPR2 mutation carriers who have not developed PAH should have annual echocardiography surveillance. Pregnancy contraindicated in PAH (maternal mortality ~30-50%).
	74	Cardiac Rhabdomyoma	TSC1 (hamartin, chromosome 9q34) or TSC2 (tuberin, chromosome 16p13.3); mTOR pathway hyperactivation	AD (TSC1/TSC2); ~50% de novo	~1 in 100,000. Present in ~50% of tuberous sclerosis complex (TSC) patients. Frequently detected on fetal echocardiography -- fetal cardiac rhabdomyoma should prompt TSC genetic evaluation of the family. Most rhabdomyomas regress spontaneously; large or multiple tumors cause obstruction, arrhythmia, or heart failure. TSC has multisystem involvement: brain hamartomas (cortical tubers, SEGAs), renal angiomyolipomas, pulmonary LAM.	Observation -- majority regress spontaneously in infancy with TSC treatment. Everolimus (mTOR inhibitor): FDA approved for TSC; dramatically reduces rhabdomyoma size and prevents new tumor formation; now standard of care for TSC-associated rhabdomyomas causing hemodynamic compromise or severe arrhythmia. Surgical resection only for life-threatening obstruction not responsive to everolimus. Genetic testing for TSC1/TSC2: mandatory in all patients with cardiac rhabdomyoma -- identifies TSC before neurological or renal manifestations emerge. EEG and brain MRI for neurological TSC surveillance.

 TIER 1 — ASSOCIATED: Germline syndrome with strong but non-universal cardiac link; syndromic screening should be considered in the appropriate clinical context

TIER	#	CONDITION	SYNDROME / GENE(s)	INHERITANCE	CLINICAL NOTES	CARDIAC KEY POINTS
	5	Pulmonary Stenosis	PTPN11/RASopathy genes (Noonan syndrome); ELN (Williams syndrome); JAG1/NOTCH2 (Alagille syndrome)	AD for each associated syndrome (most de novo)	Isolated PS in isolation is not typically a germline syndrome. However, ~25% of PS cases are associated with Noonan syndrome (dysplastic pulmonary valve type), Williams syndrome (bilateral peripheral PA stenosis), or Alagille syndrome (diffuse peripheral PA stenosis). Each association has distinct morphology and surgical implications. Noonan: dysplastic valve leaflets with myxomatous changes -- balloon valvuloplasty has lower success than typical PS.	PS management is syndrome-specific: Noonan dysplastic valves require surgical repair more often than typical PS (balloon valvuloplasty lower success). Williams syndrome peripheral PA stenosis: transcatheter stenting preferred; serial procedures throughout childhood. Alagille diffuse peripheral PA stenosis: serial catheterizations required; goal is to reduce RV pressure progressively. All three syndromes require genetic evaluation to identify associated features beyond the cardiac lesion.
	7	Atrioventricular Septal Defect (AVSD)	Trisomy 21 (Down syndrome) in 40-50%; 22q11.2 deletion in rare cases; non-disjunction	Chromosomal (Down syndrome); de novo	~3 per 10,000 live births; 4-5% of CHD. 40-50% of complete AVSD cases have Down syndrome; 60% of cardiac defects in Down syndrome are AVSD. Down syndrome-associated AVSD: often more aggressive pulmonary hypertension development; requires repair in first few months of life. Higher risk of post-operative pulmonary hypertension.	Open heart surgery (patch closure of both holes + valve reconstruction) in first months of life. Pre-operative management of pulmonary hypertension is critical in Down syndrome patients (higher PVR at baseline). Genetic testing for all AVSD patients (chromosomal microarray). Down syndrome: trisomy 21 -- guides management of associated conditions (hypothyroidism, duodenal atresia, hearing loss, leukemia risk). Post-operative pulmonary hypertension management: iNO, sildenafil, oxygen.
	8	Partial AVSD (Primum ASD)	Trisomy 21 (Down syndrome) in ~20%; less common than in complete AVSD; non-disjunction	Chromosomal (Down syndrome); de novo	~1-2 per 10,000 live births; ~30% of all AVSDs. Down syndrome association ~20% (vs ~60% in complete AVSD). Isolated cleft mitral valve without ASD: a distinct subset, even rarer -- not associated with Down syndrome. Often misclassified as simple secundum ASD	Surgical repair: single-patch closure of primum ASD + direct suture closure of left AV valve cleft. Timing: 3-6 months if MR significant; defer to 1-2 years if mild. Operative mortality <1% at experienced centers. Chromosomal microarray for all AVSD patients. Down syndrome patients require monitoring for associated

					but mitral valve pathology and surgical approach are fundamentally different.	conditions. Late MR recurrence from cleft reopening: ~5-10% requiring re-repair. Lifelong cardiology follow-up for mitral valve surveillance.
	9	Tetralogy of Fallot (TOF)	22q11.2 deletion in ~15%; Trisomy 21 in ~8%; GATA4, NKX2-5, TBX1 in rare cases	Chromosomal (22q11.2, Down); de novo; rarely inherited	3-6 per 10,000 live births; 7-10% of all CHD. ~15% of TOF cases have 22q11.2 deletion (particularly with absent pulmonary valve and pulmonary atresia variants). ~8% have Down syndrome. Remaining cases are mostly sporadic; multifactorial inheritance likely in most. 22q11.2-associated TOF: associated immune deficiency, hypocalcemia, palatal abnormalities.	Surgical repair per standard TOF protocol. 22q11.2 testing: strongly recommended in all TOF patients -- guides immune management (irradiated blood products if immunocompromised), calcium supplementation, and family counseling. Down syndrome: monitor for pulmonary hypertension (higher baseline PVR). 22q11.2 positive patients: psychiatric surveillance into adulthood (schizophrenia risk ~25%). Post-repair lifelong follow-up: MRI every 1-2 years for RV assessment, ICD risk stratification.
	15	Truncus Arteriosus	22q11.2 deletion in ~15% of cases; TBX1 haploinsufficiency	Chromosomal (22q11.2 de novo); remainder sporadic	~1 per 10,000 live births; 1-2% of CHD. 22q11.2 deletion in ~15% -- less common than in IAA (~50-70%) but clinically important to identify. The pulmonary valve pattern and morphology of truncus are associated with the 22q11 deletion in some cases.	Surgical repair in neonatal period: close VSD, separate pulmonary arteries, place RV-to-PA conduit. 22q11.2 testing recommended. Multiple conduit replacements anticipated; TPVR (Melody/SAPIEN/Harmony) defers or replaces surgical reinterventions. Branch PA stenosis at bifurcation: catheter balloon dilation and stenting. 22q11.2-positive patients: irradiated blood products, calcium supplementation, immune assessment.
	75	Cardiac Fibroma	PTCH1 gene (Gorlin syndrome / Basal Cell Nevus Syndrome) in ~3% of cardiac fibroma cases	AD (Gorlin syndrome); most cardiac fibromas are sporadic	Rare; <1 per 100,000. Second most common primary cardiac tumor in children. Benign fibrous tumor, non-regressing. Gorlin syndrome (PTCH1) associated in ~3% of pediatric cases -- multiple basal cell carcinomas, jaw keratocysts, skeletal anomalies, and brain tumors. Significant SCD risk from VT/VF with large interventricular septal involvement.	Surgical resection (partial or complete) for large symptomatic tumors. ICD for VT/SCD prevention. Catheter ablation for arrhythmia. Cardiac transplantation for unresectable tumors. PTCH1/Gorlin syndrome evaluation: consider in pediatric cardiac fibroma; if confirmed, multidisciplinary surveillance for skin, jaw, and brain tumors. Gorlin syndrome patients need annual dermatology and dental evaluation for basal cell carcinomas and jaw keratocysts.
	76	Cardiac Myxoma (Pediatric)	PRKAR1A gene (Carney complex) in ~10% of pediatric cardiac myxomas; cAMP-dependent protein kinase regulatory subunit	AD (Carney complex); most cardiac myxomas are sporadic	Rare in children; <5% of cardiac myxomas occur in those under 15. Carney complex (PRKAR1A mutation) associated in ~10% of pediatric myxomas -- characterized by cardiac myxomas (multifocal, recurrent), spotty skin pigmentation, primary pigmented nodular adrenocortical disease (Cushing syndrome), and endocrine tumors. Adult sporadic myxomas are not associated with Carney complex.	Surgical excision with wide margins. Carney complex: regular surveillance echo every 12 months LIFELONG due to high recurrence risk (20%) -- multifocal recurrences at any cardiac site. PRKAR1A genetic testing: mandatory in pediatric cardiac myxoma, bilateral myxoma, or recurrent myxoma. Carney complex evaluation: endocrinology (Cushing syndrome, acromegaly, thyroid tumors), dermatology (spotty pigmentation), ophthalmology. Anticoagulation for embolic events.